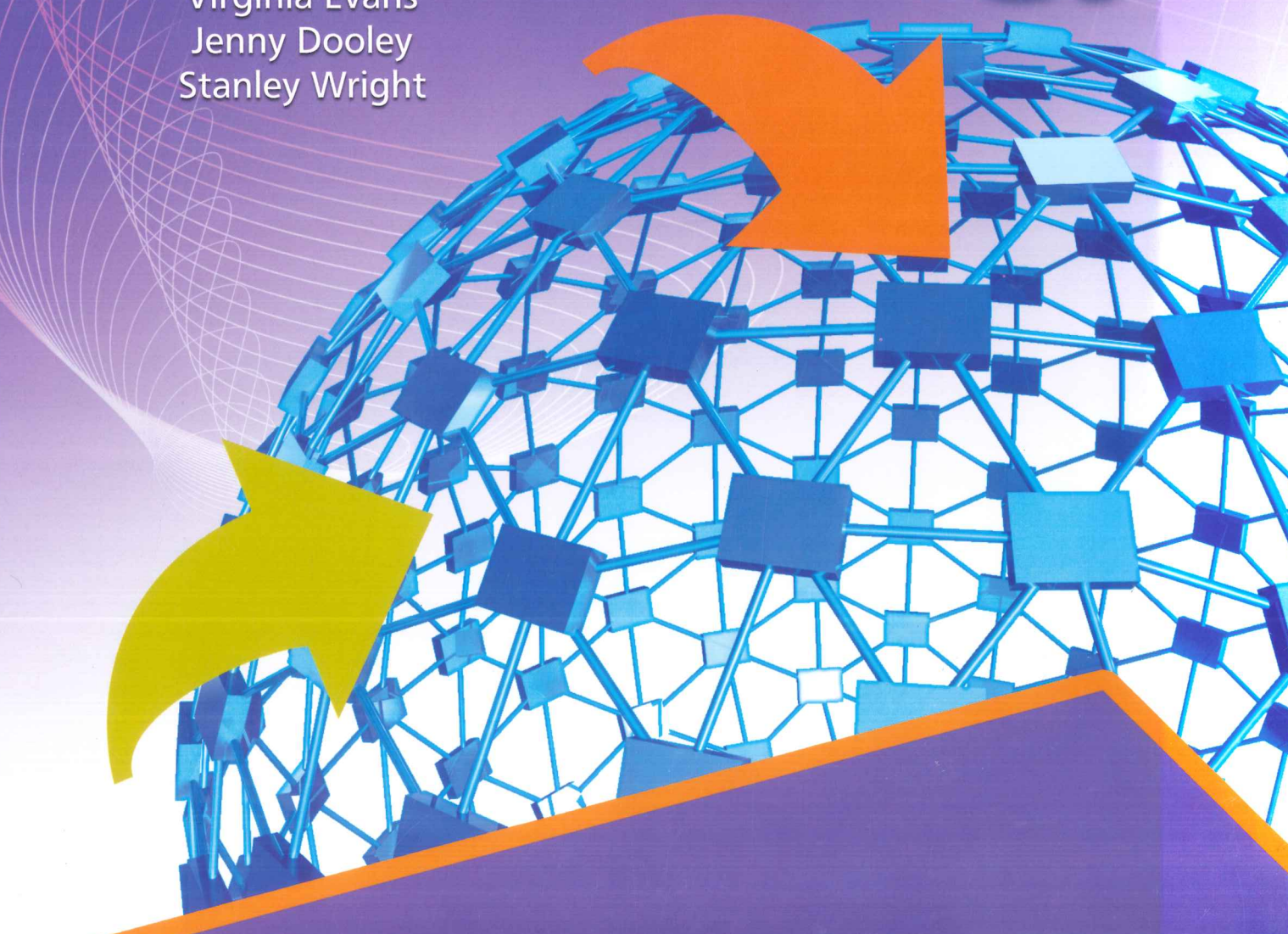
CAREER PATHS

Information Technology

Virginia Evans
Jenny Dooley
Stanley Wright



Express Publishing

TES Total Education Solutions

CAREER PATHS

Information Technology

Book 1

Virginia Evans
Jenny Dooley
Stanley Wright

CÔNG TY TNHH EDUCATION SOLUTIONS VIỆT NAM
Kính tặng

Express Publishing

Scope and Sequence

Unit	Topic	Reading context	Vocabulary	Function
1	Information Technology	Manual	code, data processing, data, encoding, hardware, information security, online, quality assurance, software, technical support	Getting someone's attention
2	Introduction to Computing Systems	Email	computer, desktop computer, laptop, mainframe, PDA, server, smartphone, supercomputer, tablet, workstation	Giving instructions
3	Inside the Computer	Product guide	case, expansion card, fan, hard drive, heat sink, motherboard, power supply, processor, RAM	Describing a process
4	Computing Devices	Catalog	input device, microphone, monitor, mouse, printer, scanner, scroll wheel, touch screen, USB	Making a suggestion
5	Networking	Magazine article	CAT-5, configuration, cycle, DHCP, Internet, IP address, LAN, network, router, WLAN	Describing options
6	The User Interface	Manual	desktop, drag, folder, GUI, icon, maximize, menu, minimize, OS, pointer, shortcut, window	Offering a solution
7	Word Processing	Advertisement	compatibility, document, font, format, formatting, layout, macro, spell check, template, word processing program	Clarifying a situation
8	Email	Memo	attachment, domain name, email address, email client, ISP, password, POP3, SMTP server, username, webmail	Checking for errors
9	Web Browsing	Email	HTML, http, hyperlink, meta tag, search engine, URL, web browser, web host, web site, www	Expressing confusion
10	Images and Graphic Design	Magazine article	contrast, crop, exposure, graphic editing program, graphics, JPEG, pixel, resize, resolution, tint	Answering the phone
11	Databases vs. Spreadsheets	Email	cell, database field, database, formula, function, operator, query, sort, spreadsheets, table, worksheet	Asking for help
12	Web Design vs. Web Development	Course descriptions	appearance, coding, content, CSS, functionality, navigation, usability, visibility, web design, web development	Asking about purpose
13	Desktop Publishing	Job listing	clip art, comprehensive layout, design, desktop publishing software, desktop publishing, electronic pages, electronic paper, electrostatic printing, graphic communications, offset lithography, page layout	Talking about experience
14	Videoconferencing	Email	bandwidth, dedicated system, document sharing, echo-cancellation, monitor, multipoint videoconference, video bridge, videoconferencing, webcam	Explaining differences
15	E-commerce	Email	affiliate, banner, brick and mortar, certificate authority, digital certificate, e-commerce, FAQs, shopping cart, traffic	Clarifying information

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2 Introduction to Computing Systems

Get ready!

1 Before you read the passage, talk about these questions.

1 What kind of computers do you use?
2 What do you use computers for at work?

From: bruce.roberts@tei.com
To: chris.carter@techmagazine.net
Re: TEI **Computers**?

Thanks for including TEI Inc. in your article on workplace technology. We have many types of computers at our disposal. First, all employees receive **desktop computers**. But we only use those for word processing and spreadsheets. Of course, some programs are too powerful for desktops. So many employees use the faster **workstations**. And we do have a few **laptops** and **tablet**. However, employees only use them on business trips. Most employees just answer email with their **smartphones**. The new **server** provides access to all printers. Finally, our **mainframe** processes our largest data files. No need for a **supercomputer** yet!

Regards,
Bruce Roberts

Reading

2 Read the email about computers available at TEI Inc. Then, choose the correct answers.

1 What is the email about?
- A the types of work employees do
- B computers that the company sells
- C what computers the company uses
- D the installation of a server

2 According to the passage, employees use laptops when they _____.
- A respond to email
- B work on spreadsheets
- C process the largest files
- D work outside of the office

3 Which of the following is NOT true?
- A The company has a mainframe.
- B Employees get email on their smartphones.
- C All employees receive desktop computers.
- D The desktop computers are faster than the workstations.

Vocabulary

3 Read the sentence and choose the correct word.

1 My **(tablet / mainframe)** notifies me when I have a meeting.
2 I have a **(laptop / supercomputer)** to do work on the plane.
3 John has a **(server / desktop computer)** because he only uses simple programs.
4 This **(computer / server)** runs programs, but doesn't have Internet access.
5 A **(workstation / tablet)** is smaller than a laptop, but still has a keyboard.
6 Connect to the **(server / laptop)** to use that printer.
7 A **(supercomputer / tablet)** is the most powerful machine in the world.

4 **Match the words (1-3) with the definitions (A-C).**

1 __ workstation
2 __ mainframe
3 __ smartphone

A large and powerful computer that supports many other computers working at once

B fast computer that is used by one person and has more memory than an ordinary personal computer.

C a device that makes and receives calls and can connect to the Internet.

5 **Listen and read the email about computers available at TEI Inc. again. Why do some employees not use a desktop?**

Listening

6 **Listen to a conversation between an IT professional and a new employee. Choose the correct answers.**

1 Which of the following computers are NOT shared by employees at TEI Inc.?

A the server
B workstations
C laptops
D desktop computers

2 According to the dialogue, employees use the server to _____.

A open programs
B reserve workstations
C log onto desktop computers
D borrow laptops

7 **Listen again and complete the conversation.**

A: Here's your new office. Jon works in here, too.
B: So, everyone has their own **1** _____ _____. Do we have laptops?
A: Yes. That's your computer there. Laptops are in the **2** _____ _____.
B: Ok, but I also need a **3** _____.
A: Right, you work on some powerful programs. Those are next door.
B: Do I sign up to use them? I mean, how do I get access to one?
A: Just **4** _____ _____ to the server. All computers connect to it. Then, click **5** _____ workstation.
B: Thanks!

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

Ok, but I also need a workstation.
How do I get access to one?
Then click 'reserve workstation'

Student A: You are an IT Professional. Answer Student B's questions about the computers.

Student B: You are a new employee. Ask Student A questions about:

- what computers they have
- where the computers are
- how to access computers

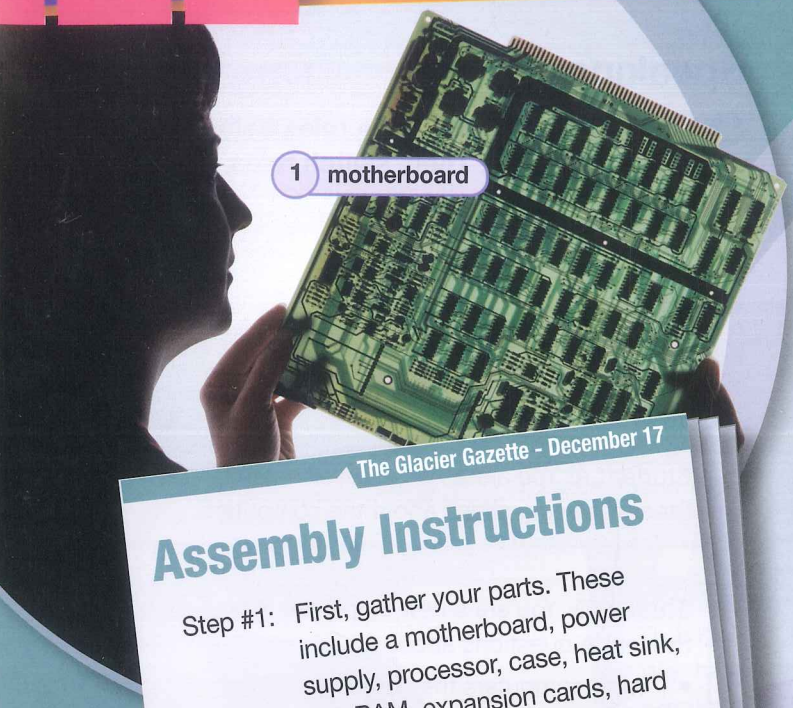
Writing

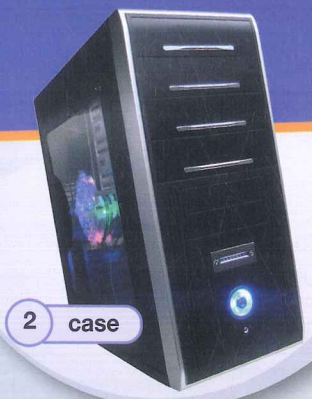
9 **Use the conversation from Task 8 to fill out the notes.**

TEI Inc.
Computers & Locations

1. _____: Every employee has one in their office.
2. _____: These are available in the IT department.
3. Server: All computers _____.
4. _____: Log on to the server to reserve these.

3 Inside the Computer





The Glacier Gazette - December 17

Assembly Instructions

Step #1: First, gather your parts. These include a motherboard, power supply, processor, case, heat sink, fan, RAM, expansion cards, hard drive, and disk drive.

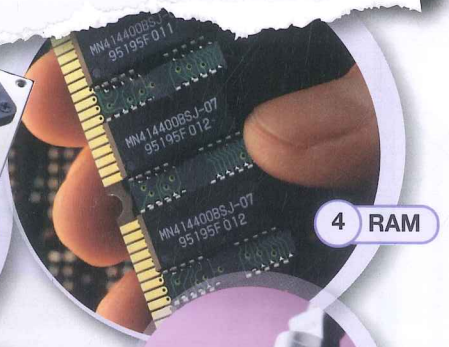
Step #2: Next, Attach the **motherboard** to the **case**. Then, connect the **processor** to the **motherboard**'s primary socket.

Step #3: Attach the **heat sink** and cooling **fan** to the processor.

Step #4: Insert the **RAM** card into an open memory slot. Put any **expansion cards** into the expansion slots.

Step #5: Connect your **hard drive** to the motherboard and **power supply**.

Step #6: Close the case and connect the power supply.



Get ready!

1 Before you read the passage, talk about these questions.

1 Which components make up a typical desktop computer?

2 What do you call a person who repairs computers?

Reading

2 Read the technology guide about computer assembly. Then, choose the correct answers.

1 What is the guide mostly about?

A putting together a computer
B problems with computer parts
C uses of expansion cards
D repairing a broken computer

2 When should the hard drive be installed?

A Step #1 **C** Step #5
B Step #3 **D** Step#6

3 According to the guide, what is NOT an assembly step?

A collecting parts
B opening the case
C connecting the drives
D installing the RAM card

Vocabulary

3 Match the words (1-6) with the definitions (A-F).

1 __ heat sink
2 __ RAM
3 __ expansion card
4 __ processor
5 __ hard drive
6 __ motherboard

A a slotted board with circuits and ports
B a short term, quick memory source
C a part used to cool the processor
D a part used to store large amounts of data
E a part used to increase a computer's functions
F a part that completes tasks for the computer

4 **Write a word that is similar in meaning to the underlined part.**

1 Spinning blades keep the computer from overheating. f _ _
2 The electrical source directs power throughout the computer. _ _ w _ r _ _ p p _ _
3 Joey purchased a protective bag for his computer. c _ _ _ _
4 Can the coded disk reader play DVDs? d _ _ _ d _ _ v _

5 **Listen and read the technology guide about computer assembly again. What should be hooked up to the motherboard?**

Listening

6 **Listen to a conversation between a technology support specialist and a customer. Mark the statements as true (T) or false (F).**

1 __ The woman replaces the disk drive.
2 __ The woman disconnects the power supply first.
3 __ The hard drive connects to the motherboard.

7 **Listen again and complete the conversation.**

Support Specialist: Technology Support, Ivan speaking. How can I help?

Customer: I have an old **1** _____ _____ and need help taking it out.

Support Specialist: Okay. Unplug the connection to the power supply first. Then, disconnect the **2** _____ cable.

Customer: Which one is the **3** _____ _____ ?

Support Specialist: It's the smaller black box in the corner.

Customer: I'm **4** _____ both. What's next?

Support Specialist: Next, take out the two small screws. They fasten the hard drive to the **5** _____ .

Customer: I see, and then it slides out. Thanks!

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

I need help taking out ...
Disconnect the ...
What is the next step?

Student A: You are a customer. Ask Student B questions about:

- taking out an old hard drive
- the power supply
- the next step in dismantling the old hard drive

Student B: You are a Support Specialist. Answer Student A's questions about dismantling the old hard drive.

Make up a name for the specialist.

Writing

9 **Use the conversation from Task 8 to fill out this feedback form. Invent names where necessary.**

Technology Support Summary

Technician: _____

Date: _____

Caller's name: _____

Problem: The caller needs to _____ an old _____ .

Suggested Action(s): _____

_____

_____

_____

_____

Was the problem resolved? **Y / N**

4 Computing Devices

COMPUTERS INTERNATIONAL — MAY ISSUE - page 14

Peripherals

Device:	Description:	Cost:
S57 Keyboard	This **input device** is available in three languages. It has a **USB** connection.	$75
Standard Mouse	Our **mouse** has a wireless connection, two buttons and a **scroll wheel**.	$64
Ep320 Scanner	This **scanner** reads documents up to 216 mm by 297 mm.	$179
PR-15 Microphone	Our **microphone** collects crystal clear sound.	$56
LCD **Monitor**	Our LCD (Liquid Crystal Display) monitors offer crisp images, and take up little space.	$167
Prinpro 2000 **Printer**	The Prinpro2000 produces color pictures and documents quickly.	$99
SoftTouch Touch Screen	The **touch screen** allows input without a keyboard and is programmable for many uses.	$280

Get ready!

1 Before you read the passage, talk about these questions.

1 How do computers receive information from users?

2 What devices can you attach to computers?

Reading

2 Read the advertisement from a computer parts catalogue. Then, mark the following answers as true (T) or false (F).

1 __ The keyboard has a wireless connection.

2 __ The LCD monitor does not require a lot of room.

3 __ The touch screen comes with a keyboard.

Vocabulary

3 Match the words (1-5) with the definitions (A-E).

1 __ keyboard
2 __ mouse
3 __ scanner
4 __ microphone
5 __ monitor

A a device that reads images and sends them to a computer

B a device that displays images

C a device that enters information into a computer

D a device that gathers sounds

E a device that is used to control a pointer or cursor

4 **Check (✓) the sentence that is true.**

1 __ **A** A keyboard reads documents.
__ **B** A USB cable connects computers and devices.

2 __ **A** A peripheral is the main part of a computer.
__ **B** A printer creates documents and pictures.

3 __ **A** An input device enters information.
__ **B** A microphone creates sounds.

4 __ **A** A monitor inputs sound.
__ **B** A touch screen does not need a mouse.

5 __ **A** A scanner displays pictures and words.
__ **B** A peripheral is a device that is added to a computer.

5 **Listen and read the advertisement from a computer catalog again. What is a USB used for?**

Listening

6 **Listen to a telephone conversation between an employee and a business owner. Check (✓) each item that the employee suggests.**

1 ❑ Monitor
2 ❑ Touch Screen
3 ❑ Keyboard
4 ❑ Mouse
5 ❑ Printer
6 ❑ Scanner

7 **Listen again and complete the conversation.**

A: Computers International, can I help you?
B: Yes. My firm's computer system needs an **1** _____.
A: What type of business is it?
B: A restaurant. Here's the problem. Customer's orders take too long to **2** _____.
A: Then, I suggest our new **3** _____ _____. You just move the data on the screen with your fingertip.
B: That sounds good. I want to create menus, too. Do you have **4** _____ for that?
A: Well, I recommend a new **5** _____. The new ones print as well as a professional service.
B: Great. Let's talk about prices.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

My computer system needs an upgrade.
What type of business is it?
I suggest our ...

Student A: You are an employee helping a business owner who needs an equipment upgrade. Ask Student B about:

- what he or she wants
- his or her type of business
- types of peripherals needed

Student B: You are a business owner. Ask the employee for help with a computer upgrade.

Writing

9 **Using the conversation from Task 8, fill in your feedback on the form.**

COMPUTERS INTERNATIONAL

Customer feedback form

1 Was our employee helpful?
Y / N

2 Did our employee make suggestions?
Y / N

3 What products were you interested in?
_____
_____

4 What peripherals did our employee recommend?
_____
_____

5 Networking

Get ready!

1 Before you read the passage, talk about these questions.

1 What are some components of a computer network?
2 How does a computer network help a business?

PC Tech Vol. 29, No. 3

Solutions for Networking

Network Troubleshooting

People often call IT professionals about **network** connection problems. But many problems are easy to fix without help. First, reconnect any loose **CAT-5** cable connections. Next, investigate **IP address** issues. This group of numbers allows communication on the **Internet**. Renewing your **DHCP** fixes most IP problems. Next, check the **router**. The router links **LANs** to the Internet. Make sure the router's **configuration** is correct. **WLANs** have other problems. Interference often blocks their wireless signals. But there is a simple solution. Just move the router away from walls and metals. Finally, **cycle** the network. Turn all equipment off for a full minute and then restart.

Reading

2 Read the article about networks. Then, choose the correct answers.

1 What is the main idea of the passage?
- A how to set up a network
- B correcting network problems
- C choosing network equipment
- D a comparison of LANs and WLANs

2 What can you infer from the passage?
- A A computer is not functioning during a cycle.
- B LAN routers do not work well near walls or metals.
- C Interference impacts a router's configuration.
- D Renewing an IP address fixes DHCP errors.

3 According to the passage, a LAN _____.
- A assigns IP addresses
- B requires a router
- C links CAT-5 cables
- D cycles a network

Vocabulary

3 Fill in the blanks with the correct words or phrases from the word bank.

word BANK

DHCP CAT-5 cable router network cycle WLAN Internet configuration

1 The _____ lets a network connect to the Internet.
2 A LAN is a type of _____.
3 Businesses communicate on the _____.
4 We do not use cables for our network. We have a _____.
5 Routers need a correct _____ to work properly.
6 Businesses with a LAN use _____ to connect computers.
7 It is sometimes necessary to _____ a network to make it work.
8 The _____ provides different IP addresses for computers.

4 **Choose the best definition for each word.**

1 router

A a part of a computer
B a wireless network
C a networking device

2 IP address

A a city-wide network
B an identification number
C a networking device

3 LAN

A a local network
B a networking device
C a wireless network

5 **Listen and read the article about networks again. What could help a user having problems with an IP address?**

Listening

6 **Listen to an IT professional talking with a customer about setting up a network. Choose the correct answers.**

1 What does the woman want to buy?

A a LAN
B a WLAN
C a router
D a CAT-5 cable

2 Why does the woman choose a wireless network?

A She does not have CAT-5 cables.
B Her computers are not close to each other.
C She does not like her present network.
D Her router doesn't support a LAN.

7 **Listen again and complete the conversation.**

IT Professional: Hi. Can I help you find something?
Customer: Oh, please. I need a **1** _____ .
IT Professional: Well, there are a few types. What kind of **2** _____ do you have?
Customer: I don't have one yet, but it's for the computers in my home.
IT Professional: Do you want a LAN or a WLAN? Some routers don't **3** _____ both types.
Customer: What's the difference?
IT Professional: A WLAN connects computers wirelessly. A LAN **4** _____ _____ with CAT-5 cables.
Customer: My computers are far **5** _____ . I'd need a WLAN.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

Can I help you find something?
Do you want a LAN or a WLAN?
A WLAN connects computers wirelessly.

Student A: You are a computer store employee. Your customer needs a router. Ask Student B:

- the equipment he or she needs
- type of network she uses

Student B: You are a customer. Answer Student A's questions to choose a router and network.

Writing

9 **Using the conversation from Task 8, fill in the network description sheet.**

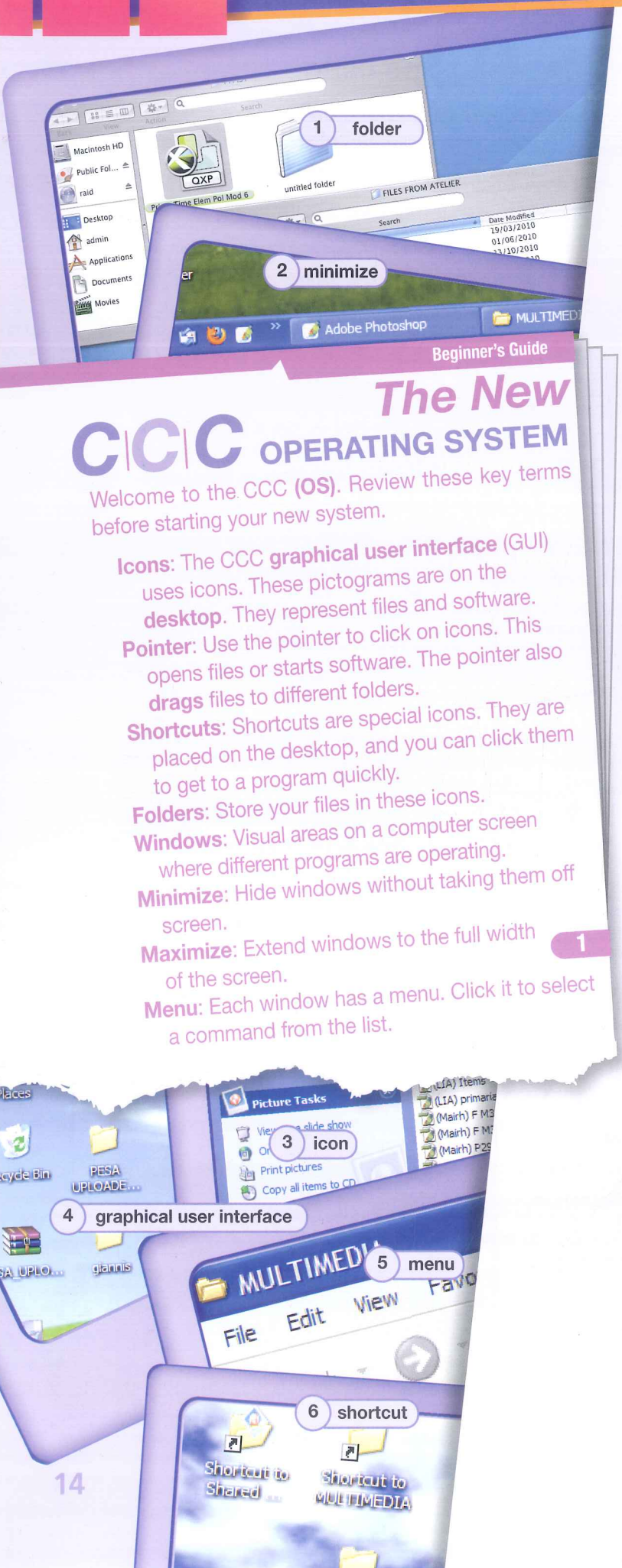
New Network Description

Location of network: _____

Type of network: _____

Components needed: _____

6 The User Interface



Get ready!

1 Before you read the passage, talk about these questions.

1 What is a User Interface?

2 How does a User Interface make computing easy?

Reading

2 Read the introduction to the operating system manual. Then, choose the correct answers.

1 What is the main idea of the passage?

A where to locate shortcuts
B the advantages of the CCC OS
C how to identify and use parts of the OS
D types of operating systems available

2 What can you infer about this operating system?

A It comes with free software.
B Files have to stay in one folder.
C Menus contain multiple commands.
D Folders do not maximize or minimize.

3 According to the passage, what does the pointer NOT do?

A move files
B open menus
C start software
D minimize windows

Vocabulary

3 Fill in the blanks with the correct words and phrases.

WORD BANK

icons desktop operating system
pointer windows GUI

1 Use the _____ to choose files and commands.

2 Small pictures or symbols in software are _____.

3 The _____ is the program that makes a computer work.

4 When something is on the _____ you see it onscreen.

5 An OS that uses small pictures to represent files is a _____.

6 You can minimize files and programs in different _____.

4 **Choose the best definition for each word.**

1 Drag
- A save an icon
- B close a desktop
- C move a file

2 Folders
- A organize files
- B move windows
- C create shortcuts

3 Shortcut
- A a type of software
- B a type of icon
- C a type of window

5 **Listen and read the introduction to the operating system manual again. Why would someone use a shortcut on their GUI?**

Listening

6 **Listen to a customer calling the help line about the operating system. Choose the correct answers.**

1 What is the dialogue mainly about?
- A Saving files
- B Organizing icons
- C Finding shortcuts
- D Locating folders

2 The customer does not know where to find the _____.
- A icons
- B desktop
- C pointer
- D options menu

7 **Listen again and complete the conversation.**

A: Hello, CCC Solutions. What do you need help with today?

B: Well, it's my **1** _____ _____. I add a shortcut one day, and it's gone the next.

A: Oh, no problem. Just **2** _____ them to the desktop.

B: Okay. How do I do that?

A: First, add the shortcuts. Then, move the pointer to the **3** _____ _____ and click 'save icons and folders'.

B: Options? **4** _____ _____, where is that?

A: See the 'CCC' icon in the corner? **5** _____ - _____ that.

B: Got it. **6** _____!

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

The shortcut is gone.

Click 'save icons and folders'.

Where is that?

Student A: You provide technical support for operating systems. Answer Student B's questions.

Student B: You need help with the new operating system. Ask Student A questions to find out how to:
- save icons
- open menus

Writing

9 **Using the conversation from Task 8, fill in the how-to sheet for the operating system.**

CCC OPERATING SYSTEM

Customer Service Call Summary Form

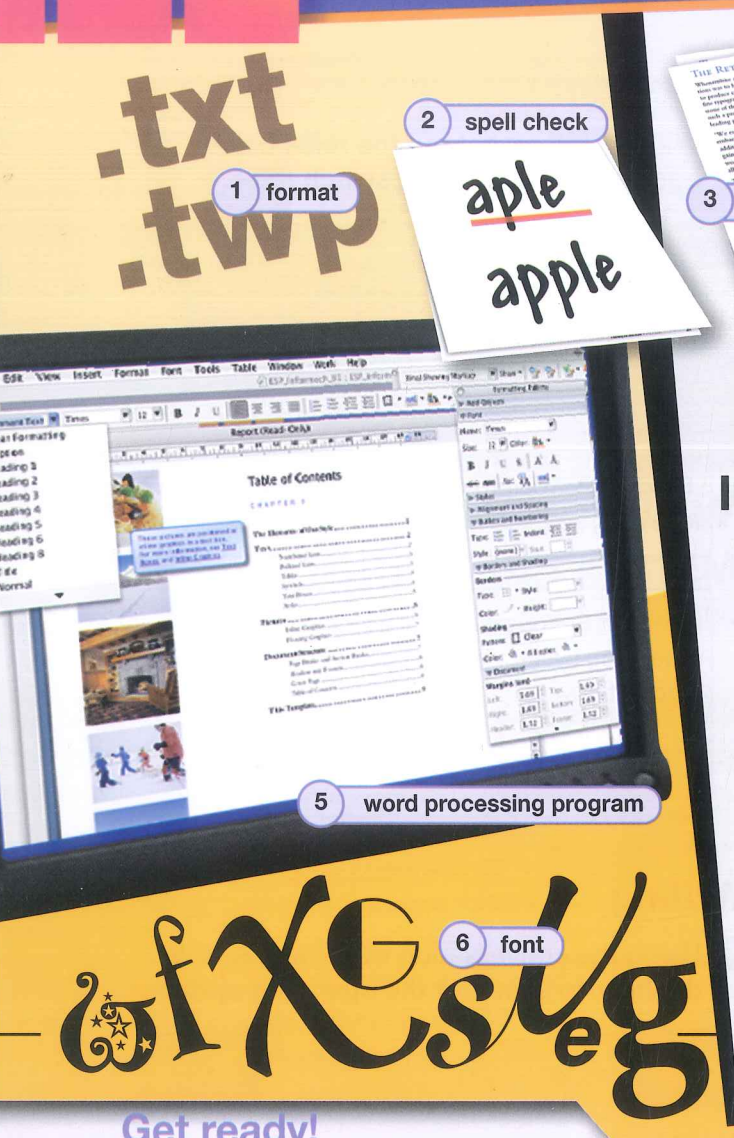
Customer Problem:
The shortcuts in the OS.

_____

Recommendation:
Move the pointer ... _____
Click on ... _____
Save the icons to ... _____

Problem Solved? **Yes / No**

7 Word Processing



INTRODUCING WORD PROCESSING TypeWrite 4.0

You need the perfect **word processing program**. And the new TypeWrite 4.0 is our best version yet. Write bold, striking **documents.** Use our many stylish **fonts**. Change your **layout** with a single click. Browse the new customizable **templates**. Customize toolbars for easier **formatting.**

It's more than a new look. TypeWrite 4.0 eliminates most **compatibility** issues. Now dozens more file **formats** work with TypeWrite. And TypeWrite 4.0 allows up to 500 **macros**, so save time as you type! TypeWrite 4.0 also features the most accurate **spell check** to date.

WORD PROCESSING TypeWrite 4.0 *Because your documents speak for you.*

Get ready!

1 Before you read the passage, talk about these questions.

1 When do people use word processing programs?

2 What kinds of word processing programs are there?

Reading

2 Read the advertisement for a new word processing program. Then, mark the following statements as true (T) or false (F).

1 __ Macros determine the layout of a document.

2 __ TypeWrite 4.0 features customizable fonts.

3 __ Not all file formats work with TypeWrite 4.0

Vocabulary

3 Fill in the blanks with the correct words and phrases from the word bank.

WORD BANK

compatibility document font format layout
macro spell check word processing program

1 Use a _____ to type out phrases that you use a lot.

2 My file won't open in other programs due to a _____ problem.

3 Instead of writing by hand, use a _____.

4 This file is in a _____ that doesn't open in TypeWrite.

5 Save each _____ with a different name so you have a copy of each.

6 The letter's _____ is wrong. The heading is in the wrong place.

7 Increase the _____ size to make the file easier to read.

8 Most word processors offer _____ to help you find mistakes.

4 **Choose the word that is closest in meaning to the underlined part.**

1 The style you use for this document makes it hard to read.
A macro B formatting C compatibility

2 To write a business letter, use the appropriate document pattern.
A word processing B compatibility C template

3 She did not think that her typing command was saving her any time.
A formatting B macro C template

5 **Listen and read the advertisement for a new word processing program again. What does the advertisement say about compatibility problems?**

Listening

6 **Listen to a conversation between an IT professional and a computer user. Choose the correct answers.**

1 What is the dialogue mostly about?
A how to open a file
B common word processing errors
C types of word processing programs
D ways to change file formats

2 What solution does the IT Professional suggest?
A buy the TypeWrite program
B download the file and try again
C get the file in a different format
D ask the client to send a program

7 **Listen again and complete the conversation.**

IT Professional: Technical Support, can I help you?
User: Yes, I have a **1** _____ _____ problem.
IT Professional: Okay. What seems to be wrong?
User: I have a file from a client, but it won't **2** _____.
IT Professional: What kind of file is it? And what program are you using?
User: It's in ".tpw" **3** _____. I use Typing Suite.
IT Professional: Oh, that's a TypeWrite file. It isn't **4** _____ with Typing Suite.
User: Oh no. I need that **5** _____.
IT Professional: Have the client send it as a ".txt" file. That opens in most **6** _____.
User: Great, thanks!

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

Can I help you?
What seems to be wrong?
That file isn't compatible.

Student A: You are an IT Professional. Ask Student B questions to find out about:
- the problem
- what program
- file format

Student B: You are a computer user. Answer Student A's questions.

Writing

9 **Use the conversation from Task 8 to fill in the email. Make up a name for the user.**

Dear _____,

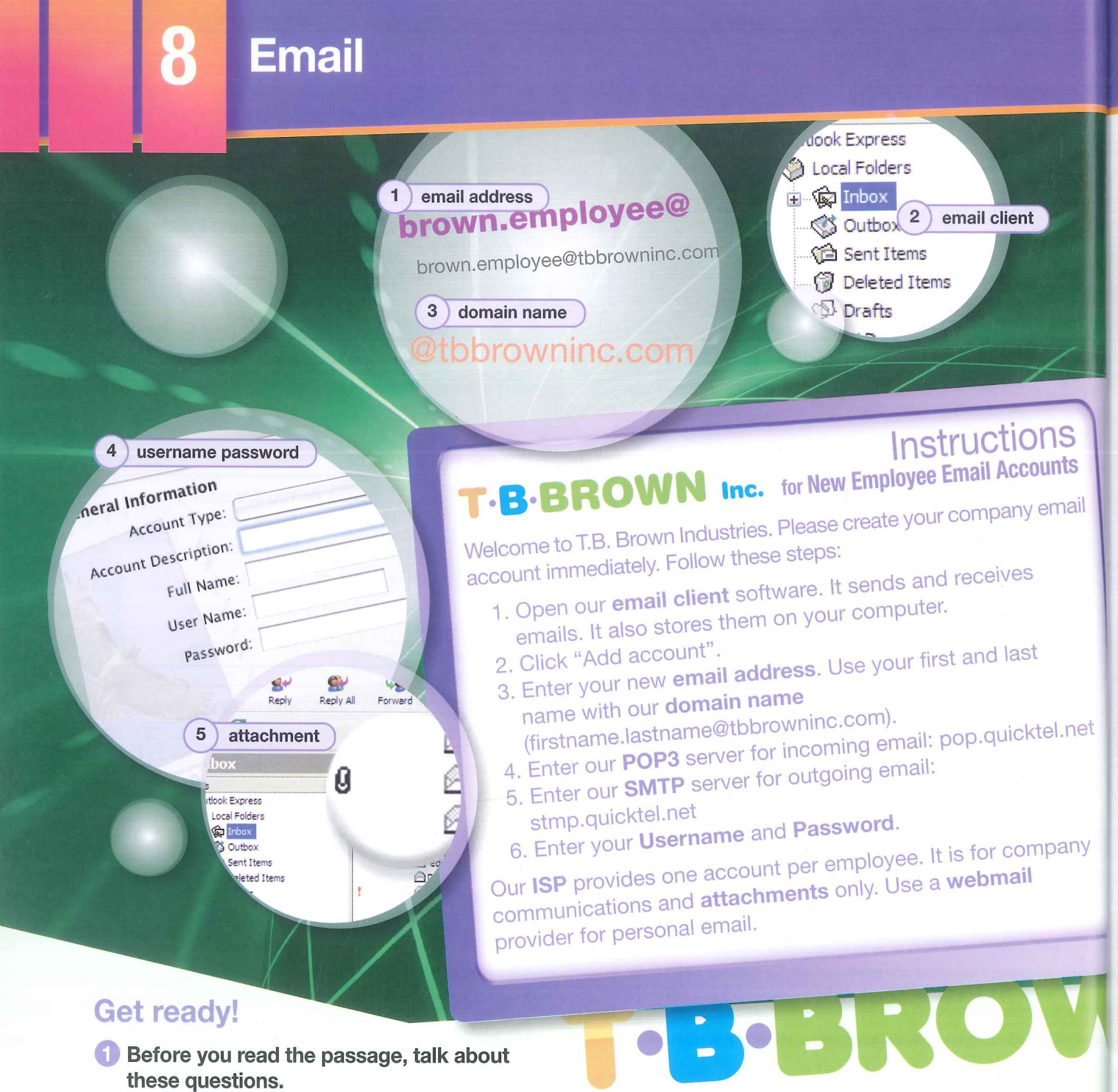
Thank you for sending the document, but _____. Your file is in the wrong _____.
My word processing program is _____. It isn't _____ with the program you're using which is TypeWrite.

Please send me the file in a .txt format. It _____ most programs.

Regards,

_____

8 Email



Get ready!

1 Before you read the passage, talk about these questions.

1 How is email helpful to people and businesses?
2 What are some problems with email?

Reading

2 Read the excerpt from the employee handbook guide at T.B. Brown Inc. Then, mark the following statements as true (T) or false (F).

1 __ The company uses a POP3 server to receive email.
2 __ The company provides webmail for all employees.
3 __ All employees have multiple company email accounts.

Vocabulary

3 Match the words (1-5) with the definitions (A-E).

1 __ attachment
2 __ domain name
3 __ password
4 __ email client
5 __ SMTP server

A a computer that sends email
B a program that saves, sends and receives email
C words/letters used to identify companies and addresses
D letters/numbers that let a person access something
E a file that is sent with an email

4 Choose the word that is closest in meaning to the underlined part.

1 That <u>company that provides access to the Internet</u> has very good service.
A ISP **B** email address **C** username

2 I use <u>an email service available on web browsers</u>, so my emails are not saved on my computer.
A POP3 **B** webmail **C** ISP

3 Enter your <u>code that identifies you on a computer</u> and press "enter".
A username **B** webmail **C** POP3

4 Most people use a <u>method to receive emails from a server</u> to get their email.
A email address **B** POP3 **C** webmail

5 Listen and read the excerpt from the employee handbook guide at T.B Brown Inc. again. What must be included with the user name when setting up the email address?

Listening

6 Listen to a conversation between a QuickTel employee and a customer. Choose the correct answers.

1 What are the speakers mostly talking about?
A how to create a new email address
B why a customer is not pleased with QuickTel
C when to enter a username and password
D what's wrong with a customer's email

2 Which of the following is NOT a potential cause of the customer's problem?
A username **C** POP3 settings
B password **D** webmail account

7 Listen again and complete the conversation.

E: Thanks for calling QuickTel. How may I help you?
C: Well, I'm a new customer, and my **1** _____ _____ isn't receiving mail.
E: Okay. Do you have the correct **2** _____ and **3** _____ ?
C: Yes. Those work fine.
E: Okay, do you have a **4** _____ account or an email client? There are some common mistakes with those.
C: An **5** _____ _____ . Does that matter?
E: New customers often make mistakes with the **6** _____ settings. The correct server is POP.QuickTel.net.
C: Oh, that's it. I have Pop.QuickTel.com. Thanks.

Speaking

8 With a partner, act out the roles below based on Task 7. Then, switch roles.

USE LANGUAGE SUCH AS:

My email address isn't receiving mail.

Do you have the correct username and password?

Does that matter?

Student A: You are a QuickTel employee. A customer has a problem with his or her email. Find the problem. Ask Student B about their:

- username
- password
- POP3 server

Student B: You are a customer. Answer Student A's questions.

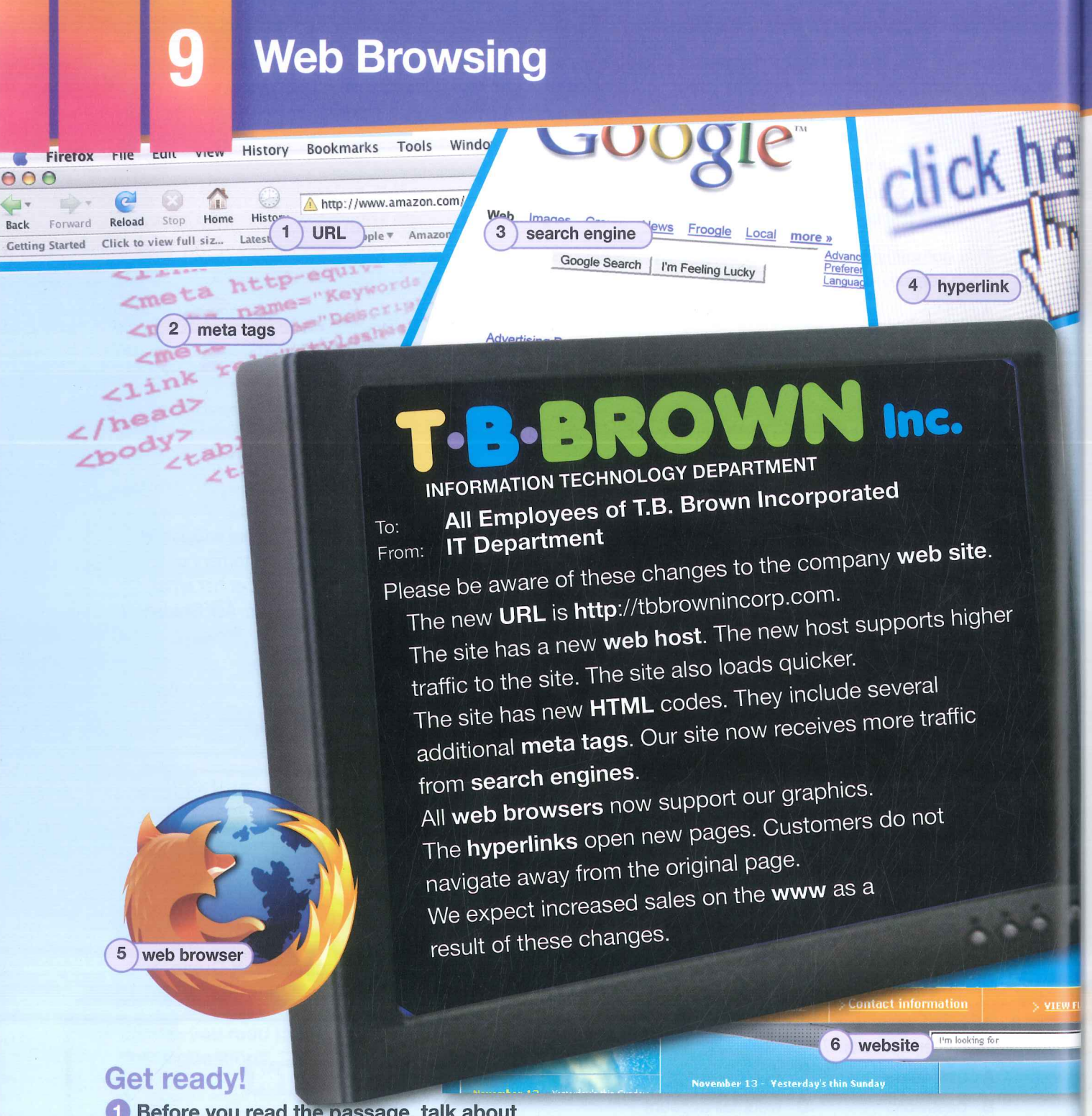
Writing

9 Use the conversation from Task 8 to complete the Troubleshooting Guide.

QUICKTel Email Troubleshooting Guide

1 Is the _____ correct? **Y / N**
2 Is the _____ correct? **Y / N**
3 What _____ is the customer using?
4 Does that server name match ours? **Y / N**
5 Problem resolved by: _____

9 Web Browsing



Get ready!

1 Before you read the passage, talk about these questions.

1 When do you browse the web at work?
2 What kind of things do you browse for?

Reading

2 Read the email from the IT department of T.B. Brown Inc. Then, mark the following statements as true (T) or false (F).

1 __ T.B. Brown Inc. now has a different URL.
2 __ The hyperlinks navigate away from the original page.
3 __ The web site now has fewer meta tags.

Vocabulary

3 Match the words (1-5) with the definitions (A-E).

1 __ hyperlink
2 __ website
3 __ HTML
4 __ web host
5 __ web browser

A a collection of related Internet pages
B a program that lets you look at web pages
C a word/image that you can click on to navigate
D a computer language used to make web sites
E a company that provides space for a website

4 Choose the word that is closest in meaning to the underlined part.

1 The text used in webpage codes that provide information to search engines gives us high search rankings.
A hyperlink B meta tag C web host

2 Type a word into the computer program that allows people to look for particular information to find a web site.
A search engine B http C website

3 Our website is very popular on the system of information that is available on the Internet.
A meta tag B hyperlink C www

4 The company's text and symbols that form a web address is http://tbbrownincorp.com.
A URL B search engine C web host

5 Please begin the web address with the part of the web address that allows the computer to connect to the site.
A HTML B web browser C http

5 Listen and read the email from the IT department of T.B Brown Inc. again. Where does their site get more activity from?

Listening

6 Listen to a conversation between two employees of T.B. Brown Inc. Choose the correct answers.

1 What is the main idea of the dialogue?
A how to compose a memo
B the importance of meta tags
C finding a better search engine
D what a hyperlink is

2 What does the man say about the website?
A It displays larger meta tags.
B It contains more hyperlinks.
C It has larger web pages available.
D It appears near the top of search results.

7 Listen again and complete the conversation.

B: Hi Susan! Did you receive the staff email?
S: Yes, and I have a question, Bob.
B: Okay, no problem. What is it?
S: I don't even know what **1** _____ _____ are. Why are they important?
B: They're descriptive words on the **2** _____ _____. They make the site rank higher in **3** _____ _____.
S: You mean our **4** _____ appears at the top of the search results?
B: Exactly. Customers **5** _____ _____ our hyperlink first.
S: That's good news for our sales department.

Speaking

8 With a partner, act out the roles below based on Task 7. Then, switch roles.

USE LANGUAGE SUCH AS:

I don't even know what meta tags are.
Our site appears at the top of the search results.
Customers click on our hyperlink first.

Student A: You are a T.B. Brown Inc. employee. You want to know more about meta tags and the company's website. Talk about:
- meta tags
- search engines
- search results

Make up a name for your co-worker.

Student B: You are an IT department employee. Answer A's questions.

Make up a name for your co-worker.

Writing

9 Use the conversation from Task 8 to fill in the email.

Dear _____,
Thanks for explaining what the _____ are.
I had no idea they were so _____. Now,
I can see why our website appears at _____.
There are some pretty good _____
on the new website. Anyway, as long as customers keep _____ first, it'll be good for business!
Regards,
_____

Glossary

affiliate [N-COUNT-U15] An affiliate is a person or business that directs Internet traffic to a business's website in exchange for a percentage of sales.

appearance [N-COUNT-U12] Appearance is the way that a web page looks.

attachment [N-COUNT-U8] An attachment is a file that is sent with an email.

bandwidth [N-UNCOUNT-U14] Bandwidth is a measurement of the amount of information that a computer network can send or receive. High bandwidth allows images and websites to load quickly on the Internet.

banner [N-COUNT-U15] A banner is a rectangular graphic on a website that advertises a business.

brick and mortar [N-COUNT-U15] A brick and mortar is a business that operates in a physical location instead of online.

case [N-COUNT-U3] A case is an enclosure that holds the computer's components.

CAT-5 (Category 5) cables [N-COUNT-U5] CAT-5 cables are the standard cables that connect computers to networks.

cell [N-COUNT-U11] A cell is a unit of a spreadsheet that holds a piece of information.

certificate authority [N-COUNT-U15] A certificate authority creates, offers and verifies reliable digital certificates.

clip art [N-UNCOUNT-U13] Clip art is electronic artwork that can be used in publications.

code [N-COUNT-U1] Code is a set of words and signs that give instructions to a computer.

codec [N-COUNT-U14] A codec is a piece of videoconferencing equipment. It codes signals that go out and reads signals that come in.

coding [N-UNCOUNT-U12] Coding is the programming language that makes up a website.

compatibility [N-UNCOUNT-U7] Compatibility is the ability of one program or file to work with a different program.

comprehensive layout [N-UNCOUNT-U13] A comprehensive layout is a nearly complete version of a publication.

computer [N-COUNT-U2] A computer is an electronic device that manipulates data.

configuration [N-COUNT-U5] The configuration of a network is the group of settings that control how information is sent and received on a network.

content [N-UNCOUNT-U12] Content is the material on a web page, including text and graphics.

contrast [N-UNCOUNT-U10] Contrast is the amount of difference between the dark areas and light areas of an image.

crop [V-COUNT-U10] To crop is to cut off unwanted parts of an image.

CSS [N-UNCOUNT-U12] CSS (cascading style sheets) is a simple language that decides how a web page looks.

cycle [V-UNCOUNT-U5] To cycle a device on a network means to turn it off and then turn it on again or to unplug it from its power source.

data [N-UNCOUNT-U1] Data are numbers that can be entered into a computer.

data processing [N-UNCOUNT-U1] Data processing is the act of entering, using or manipulating data with computers.

database [N-COUNT-U11] A database is a large group of data organized in a computer system. It allows you to search and find information easily.

dedicated system [N-COUNT-U14] A dedicated system includes all the necessary components of videoconferencing together as a set.

desktop [N-COUNT-U6] A computer desktop is everything that a user sees on a computer screen.

desktop computer [N-COUNT-U2] A desktop computer is a computer that is small enough to use at a desk, but too big to carry around.

desktop publishing software [N-UNCOUNT-U13] Desktop publishing software is a computer program that is used to make page layouts.

desktop publishing [N-UNCOUNT-U13] **Desktop publishing** is art of using computers to design finished, printable documents.

DHCP (Dynamic Host Configuration Protocol) [N-COUNT-U5] The **DHCP** is a system for computer networks to receive IP addresses and other configuration information.

digital certificate [N-COUNT-U15] A **digital certificate** is an electronic document that proves an online business or person is who that business or person claims to be.

disk drive [N-COUNT-U3] A **disk drive** reads information from a magnetic spinning disc.

document [N-COUNT-U7] A **document** is a file created on a word processing program.

document sharing [N-UNCOUNT-U14] **Document sharing** is a feature of some videoconferencing systems. It allows everyone participating in the videoconference to look at and edit the same document.

domain name [N-COUNT-U8] A **domain name** is a set of words and letters that identify a company or address on the Internet.

drag [V-UNCOUNT-U6] To **drag** files, folders or icons is to move them around on a desktop.

e-commerce [N-UNCOUNT-U15] **E-commerce** is the purchase or sale of items on the Internet.

echo-cancellation [N-UNCOUNT-U14] **Echo-cancellation** is a process that reduces the amount of echo heard by participants of a videoconference.

electronic page [N-COUNT-U13] An **electronic page** is a computerized version of a piece of paper.

electronic paper [N-COUNT-U13] **Electronic paper** is a special digital screen that can be read like paper.

electrostatic printing [N-UNCOUNT-U13] **Electrostatic printing** is a method that uses the forces of electricity to print.

email address [N-COUNT-U8] An **email address** is a location to which electronic messages can be delivered.

email client [N-COUNT-U8] An **email client** is a program that receives, sends and stores email on a computer.

encoding [N-UNCOUNT-U1] **Encoding** is the act of changing data so that no one can read it until it is changed again.

expansion cards [N-COUNT-U3] **Expansion cards** are extra circuit boards that are used to increase the functions of a computer.

exposure [N-COUNT-U10] An **exposure** is the length of time film is open to light when taking a picture.

fan [N-COUNT-U9] A **fan** moves cool air onto computer components inside the case.

FAQs [N-COUNT-U15] **FAQs** (Frequently Asked Questions) are a set of questions and answers that are provided on a website to provide users with information that users commonly want to know.

field [N-COUNT-U11] A **field** is a category in a database that holds a particular type of information.

folder [N-COUNT-U6] A **folder** holds and organizes files and documents on a computer.

font [N-COUNT-U7] **Font** is the style of the typed characters on a document.

format [N-COUNT-U7] A **format** is a file's type. Different files might not work in different programs.

formatting [N-UNCOUNT-U7] **Formatting** is the design of the words on a document.

formula [N-COUNT-U11] A **formula** is any mathematical calculation that you perform in a spreadsheet.

function [N-COUNT-U11] A **function** is a mathematical instruction that performs a specific calculation in a spreadsheet, such as adding a set of values.

functionality [N-COUNT-U12] **Functionality** is the ability of different elements of a website to work together.

graphic communications [N-UNCOUNT-U13] **Graphic communications** is the study of sending and receiving visual messages.

Glossary

graphic editing program [N-COUNT-U10] A **graphic editing program** is a computer program that lets you change different aspects of an image.

Graphical User Interface [N-COUNT-U6] A **graphical user interface** (GUI) is an operating system that uses icons to represent programs and files and allow users to access to them.

graphics [N-COUNT-U10] **Graphics** are images on a computer, in a book or magazine, etc.

hard drive [N-COUNT-U3] A **hard drive** stores data electronically.

hardware [N-COUNT-U1] **Hardware** refers to the physical parts of a computer.

heat sink [N-COUNT-U3] A **heat sink** transfers heat away from nearby components of a computer.

HTML [N-UNCOUNT-U9] **HTML** (Hypertext Markup Language) is a computer language that is used to construct web sites on the Internet.

http [N-UNCOUNT-U9] An **http** is a part of an Internet address that allows a computer to connect to the site.

hyperlink [N-COUNT-U9] A **hyperlink** is a word or image on a web page that you can click on to go to a different web page.

icon [N-COUNT-U6] An **icon** is a picture that represents something on a computer.

information security [N-COUNT-U1] **Information security** is the act or process of keeping information safe and preventing others from seeing it.

Internet [N-UNCOUNT-U5] The **Internet** is a worldwide network of computers.

IP (Internet Protocol) address [N-COUNT-U5] An **IP** address is a group of numbers that identify a computer on a network.

ISP [N-COUNT-U8] An **ISP** (Internet Service Provider) is a company that gives its customers access to the Internet, email addresses and data storage.

JPEG [N-UNCOUNT-U10] **JPEG** (Joint Photographic Experts Group) is a kind of computer file used it to store pictures on a computer.

keyboard [N-COUNT-U4] A **keyboard** is a set of buttons that is used to enter information into a computer.

laptop [N-COUNT-U2] A **laptop** is a computer that is small enough for you to carry around and use anywhere.

layout [N-COUNT-U7] **Layout** is the way that items are arranged on a document.

LCD [N-COUNT-U4] An **LCD** (Liquid Crystal Display) is a type of display that is created by liquid crystals and that is used in thin monitors.

local area network (LAN) [N-COUNT-U5] A **local area network** is a network that connects a small group of computers.

macro [N-COUNT-U7] A **macro** is a short keystroke that performs a predetermined action.

mainframe [N-COUNT-U2] A **mainframe** is a very powerful computer that can process large amounts of data.

maximize [V-UNCOUNT-U6] To **maximize** a window is to make it fill an entire computer screen or desktop.

meta tag [N-COUNT] A **meta tag** is text that is used in the codes of a web page to provide information about the site to search engines.

microphone [N-COUNT-U4] A **microphone** is a device that is used to record sound.

minimize [V-UNCOUNT-U6] To **minimize** a window is to make it leave the screen without closing the window.

monitor [N-COUNT-U4-U14] A **monitor** is a screen that displays text, video, or images.

motherboard [N-COUNT-U3] A **motherboard** is a firm slotted board onto which computer circuitry is attached.

mouse [N-COUNT-U4] A **mouse** is a small device that is used to move the cursor and select items on a computer screen.

multipoint videoconference [N-COUNT-U14] A **multipoint videoconference** is a videoconference among people in three or more different locations.

navigation [N-COUNT-U12] **Navigation** is the act of using and finding things on a website.

network [N-COUNT-U5] A **network** is a group of computers that are connected in order to share data or communicate.

offset lithography [N-UNCOUNT-U13] **Offset lithography** is a common form of printing.

online [ADJ-COUNT-U1] If a computer is **online**, it is connected to the Internet.

operating system (OS) [N-COUNT-U6] The **operating system** is the software that controls a computer.

operator [N-COUNT-U11] An **operator** is a symbol in a formula that performs a particular calculation, such as a plus sign (+).

page layout [V-INT-U13] **Page layout** is the process of arranging text and graphics on a page.

page view [N-COUNT-U15] A **page view** is a request from a computer to load a page of a web site. They can be counted to analyze the number of people viewing a website.

password [N-COUNT-U8] A **password** is a set of letters and/or numbers that allow someone to use or access something on a computer.

peripheral [N-COUNT-U4] A **peripheral** is a device that can be connected to a computer.

pixel [N-COUNT-U10] A **pixel** is one of the small dots that make up an image on a computer or television screen.

pointer [N-COUNT-U6] A **pointer** is a small arrow that lets a user choose what to do on a computer.

POP3 [N-COUNT-U8] A **POP3** (Post Office Protocol) is a method that allows computer users to receive email from a server.

power supply [N-COUNT-U3] A **power supply** delivers electricity to all parts of a computer.

printer [N-COUNT-U4] A **printer** is a device that is used by a computer to transfer ink onto paper.

processor [N-COUNT-U3] A **processor** carries out the instructions of computer programs.

quality assurance [N-UNCOUNT-U1] **Quality assurance** is the act of checking products for problems.

query [N-COUNT-U11] A **query** is a search that locates all information of a specific type in a database.

RAM [N-UNCOUNT-U3] **RAM** (Random Access Memory) is memory that can be quickly accessed.

resize [V-COUNT-U10] To **resize** is to make an image on a computer bigger or smaller.

resolution [N-UNCOUNT-U10] **Resolution** is the ability of a computer or camera to produce a clear and detailed image.

router [N-COUNT-U5] A **router** is a device that allows more than one computer to connect to a network at the same time.

scanner [N-COUNT-U4] A **scanner** is a device that is used for transferring printed documents and pictures into a computer system.

search engine [N-COUNT] A **search engine** is a computer program that allows people to search for particular information, products, etc. on the Internet.

server [N-COUNT-U2] A **server** is a computer that connects many computers to hardware.

shopping cart [N-COUNT-U15] A **shopping cart** is a part of an online store that displays items that a consumer has selected for purchase, but has not yet paid for.

shortcut [N-COUNT-U6] A **shortcut** is an icon that has been moved to a desktop to make opening a file or program easier.

Glossary

smartphone [N-COUNT-U2] A **smartphone** is a phone that you take with you to make calls and access the Internet anywhere.

software [N-UNCOUNT-U1] **Software** is a program or instruction that makes a computer perform a specific task.

sort [V-INT-U11] To **sort** is to arrange the information in a database in a specific way.

spell check [N-UNCOUNT-U7] **Spell check** is a function that reviews a document for spelling errors.

spreadsheet [N-COUNT-U11] A **spreadsheet** is a computer program that organizes information and performs calculations.

STMP [N-COUNT-U8] A **SMTP** (Simple Mail Transfer Protocol) is a method that allows computer users to send emails to other users.

supercomputer [N-COUNT-U2] A **supercomputer** is a large computer that is built to process large amounts of information at a higher speed than other computers can handle.

table [N-COUNT-U11] A **table** is a single collection or arrangement of information in a database.

tablet [N-COUNT-U2] A **tablet** is a computer that is smaller than a laptop with a touch screen and used to browse the Internet and check email.

technical support [N-UNCOUNT-U1] **Technical support** is the task of helping people use and understand technology.

template [N-COUNT-U7] A **template** is blank layout for certain types of documents.

tint [N-COUNT-U10] A **tint** is a small amount of a color in something.

touch Screen [N-COUNT-U4] A **touch screen** is a monitor that lets users interact with the computer by touching the screen.

traffic [N-UNCOUNT-U15] **Traffic** is the total number of users that go to and view a website in a certain amount of time.

URL [N-COUNT-U9] A **URL** (Uniform Resource Locator) is the text and symbols that form a web address.

usability [N-UNCOUNT-U12] **Usability** is how easy it is for a person to use a website.

USB [N-COUNT-U4] **USB** (Universal Serial Bus) is a commonly used type of connection or port to connect a peripheral to a computer.

username [N-UNCOUNT-U8] A **username** is a name or code that identifies a person on a computer and allows that person to access his or her files.

video bridge [N-COUNT-U14] A **video bridge** is a device that allows you to hold multipoint videoconferences. It connects calls from several different locations.

videoconference [N-COUNT-U14] A **videoconference** is a meeting or conversation between two or more people in different locations using computer technology.

visibility [N-UNCOUNT-U12] **Visibility** is how easily people find a website using search engines, based on how many other websites link to it.

web browser [N-COUNT-U9] A **web browser** is a computer program that allows people to look at web pages on the Internet.

web design [N-UNCOUNT-U12] **Web design** is the act or practice of determining how a website looks.

web development [N-UNCOUNT-U12] **Web development** is the act or practice of determining how a website works.

web host [N-COUNT-U9] A **web host** is a company that provides the space for a web site on the Internet.

web site [N-COUNT-U9] A **web site** is a collection of related pages on the Internet that contains information, pictures, etc. and is usually published by a person, group, or company.

webcam [N-COUNT-U14] A **webcam** is a small camera that connects to a computer. It broadcasts sound and video on the Internet.

webmail [N-UNCOUNT-U8] **Webmail** is an email service that people access through a web browser, and which sends and receives email but does not store it on computers.

window [N-COUNT-U6] A **window** is a rectangle on the desktop of a computer monitor that displays a file or program. Its contents are independent of the desktop screen.

wireless Local Area Network (WLAN) [N-COUNT-U5] A **wireless local area network** is a network that connects computers to a network without cables.

word processing program [N-COUNT-U7] People use **word processing programs** to write on computers.

worksheet [N-COUNT-U11] A **worksheet** is a single page of a spreadsheet.

workstation [N-COUNT-U2] A **workstation** is a very fast computer that runs powerful work programs.

www [N-UNCOUNT-U9] The **www** (World Wide Web) is the system of web pages, information, pictures, etc. that is available on the Internet.

CAREER PATHS

Information Technology

Book 2

Virginia Evans
Jenny Dooley
Stanley Wright

Express Publishing

Scope and Sequence

Unit	Topic	Reading context	Vocabulary	Function
1	Computer Memory	Guide	bit size, bit, bus speed, DIMM, hard drive, MB, MHz, motherboard, paging file, RAM, SIMM, virtual memory	Describing minimum requirements
2	Programming Languages	Letter	Assembler, BASIC, C#, C, C++, compiler, computer language, CPU, Java, linker, machine code, multi-thread	Expressing a concern
3	ISPs and Internet Access	Business letter	antivirus software, broadband, DSL, hotspot, ISP, T-1, T-3, upgrade, Wi-Fi, wireless router	Describing benefits
4	Storage Devices	Suggestion form	array, CD-R, CD-RW, data compression, failure, magnetic tape, memory storage density, RAID, SATA, USB flash drive	Asking about deadlines
5	Peripherals	Advertisement	automatic document feeder, calibration, CCD, digital camera, digital zoom, DPI, flash memory, inkjet printer, laser printer, megapixel, optical zoom, transparency adapter	Making a recommendation
6	MP3 Players	Announcement	AAC, bitrate, capacity, decode, dock, driver, freeze, gigabyte, playlist, upload	Investigating a problem
7	Cell Phones	Advertisement	4G network, apps, Bluetooth, call waiting, caller ID, cell phone, lithium-ion battery, ringtone, text message, touchscreen, voicemail	Offering help
8	Communications	Instant messages	barcode, data plan, email, EPC, instant message, scan, smartphone, spam, tablet, text, UPC	Assigning a task
9	Entertainment and Social Networking	Memo	avatar, blog, feed, forum, gaming, handle, link, MMORPG, post, profile, social networking, virtual world	Giving a summary
10	GPS Systems	Advertisement	accuracy, acquisition time, cartography, coordinates, GPS, integrity, latitude, longitude, navigation, waypoint	Describing capabilities
11	Banking	Website	account activity, account alert, bank fraud, bill pay, electronic statement, identity theft, lockout, online banking, PIN, site key, SSL	Reporting a problem
12	Education and Research	Advertisement	boolean operator, catalog, ebook, ejournal, EZproxy, index, license restriction, media, remote access, subscription, VPN	Recognizing a cause
13	Robotics	Email	actuator, autonomous, drive power, dynamic model, enabling device, enabling device, end-effector, equality constraint, joints, manipulator, pendant, robotics, serial robot	Citing a previous example
14	Electronic Publishing	Website	active matrix display, bitmapped graphics, black and white, download, ergonomic, offline, PDF reader, publication, reflow, sub-pixel rendering, system requirements, upload	Describing completion
15	Artificial Intelligence	Magazine Article	agent, algorithm, case-based reasoning, cognitive scientist, computer vision, heuristic, machine learning, model-based reasoning, pattern recognition, robot	Asking for clarification

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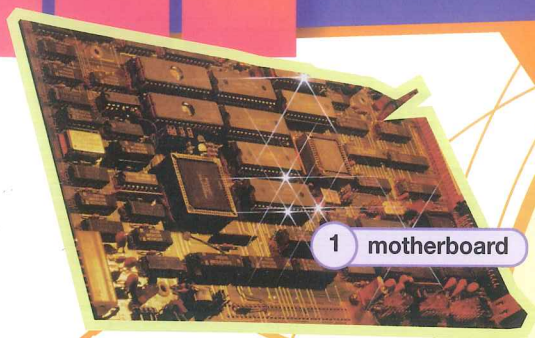
1 Computer Memory



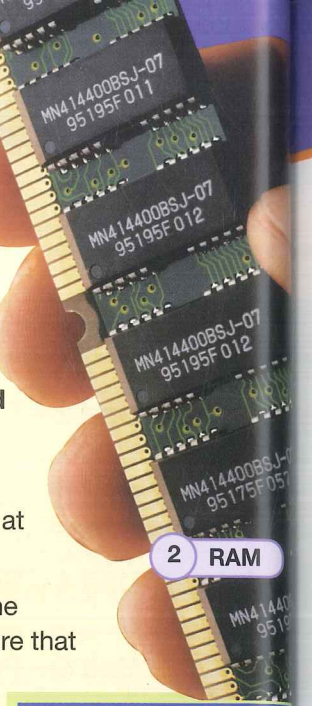
ClearPic Troubleshooting Guide

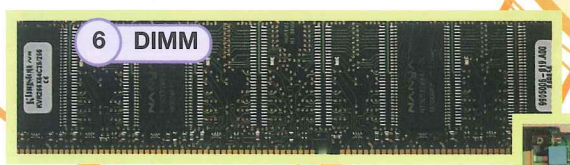
Certain memory requirements are necessary to install and run ClearPic. Check that the computer has at least 500 **MB** of **RAM** available. For optimal performance, make sure the computer's CPU has a **bit size** of at least 32 **bits**. It should run at a **bus speed** of 800 **MHz**.

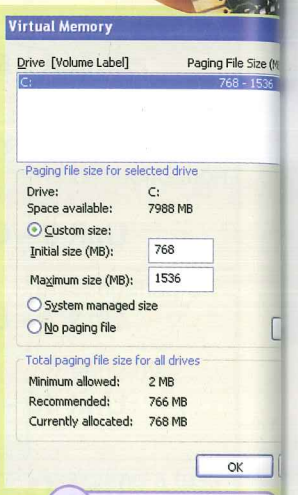
There are several options to fix problems with limited memory. One, is to limit the number of programs running at the same time.

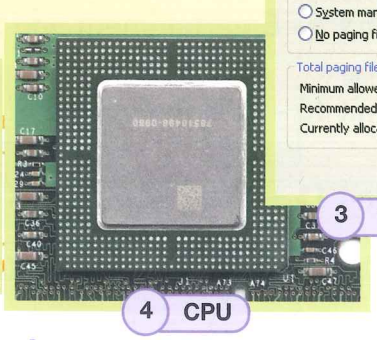
Another, is to increase the amount of RAM. First, determine whether your computer uses **SIMMs** or **DIMMs**. Make sure that the memory's **bus speed** matches that of the computer. Purchase the appropriate amount of RAM and attach it to the **motherboard**.

You can also increase your computer's **virtual memory**. This temporarily stores data on a **paging file** on the computer's **hard drive**. To add virtual memory, click on the "My Computer" icon. Under "Properties," increase the maximum memory amount.









Get ready!

1 Before you read the passage, talk about these questions.

1 How has increased computer memory changed business?

2 When do people need to add more computer memory?

Reading

2 Read the troubleshooting guide. Then, choose the correct answers.

1 What is the passage mainly about?

A the main function of the CPU
B where to install the ClearPic file
C how to add or modify computer memory
D speeds of RAM attached to the motherboard

2 What should people do first when increasing available RAM?

A check to see if they need SIMMs or DIMMs
B increase the amount of virtual memory
C temporarily store their data on a paging file
D make sure the CPU runs at a speed of 800 MHz

3 What is NOT a way to fix a limited memory problem?

A increase the amount of virtual memory
B buy a new motherboard for the computer
C reduce the number of programs running
D increase the amount of available RAM

Vocabulary

3 Match the words (1-7) with the definitions (A-G).

1 __ MHz
2 __ SIMM
3 __ motherboard
4 __ DIMM
5 __ virtual memory
6 __ paging file
7 __ bit

A where data that doesn't fit in the RAM is stored
B the main circuit board in a computer
C a unit of measurement that measures the speed of a CPU
D a data storage method that stores data on the hard drive
E the smallest unit of computer data
F an memory module that sends up to 32 bits of data to a CPU
G a memory module that sends up to 64 bits of data to a CPU

4 **Fill in the blanks with the correct words and phrases from the word bank.**

WORD BANK

hard drive RAM bit size MB bus speed

1 The computer has a _____ of 64 bits.
2 The computer's CPU has a _____ of 800 MHz.
3 Increase the virtual memory to store data on the _____.
4 This computer comes with 900 _____ of RAM.
5 Install additional _____ into the motherboard.

5 **Listen and read the troubleshooting guide again. What happens when a computer's virtual memory is made bigger?**

Listening

6 **Listen to a conversation between a customer and an employee of ClearPic. Mark the statements as true (T) or false (F).**

1 __ The customer's CPU is too slow to run the program.
2 __ The employee advises the customer to buy more RAM.
3 __ Increasing the virtual memory will increase the paging file size.

7 **Listen again and complete the conversation.**

Employee: Well, **1** _____ _____ _____. To run ClearPic you really want to have at least 500 MB of memory.
Customer: So how can I fix that?
Employee: One easy way is to increase your computer's **2** _____ _____.
Customer: What will that do?
Employee: It increases the paging file size. That way, your computer's **3** _____ _____ can store more data.
Customer: I see. So **4** _____ _____ do I increase the memory?
Employee: **5** _____ _____ on the 'My Computer' icon. Under 'Properties', increase the maximum memory amount **6** _____ _____ _____ half.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

How can I help you?
The program keeps freezing.
How can I fix that?

Student A: Your computer program keeps freezing. Talk to Student B about:

- the problem
- solutions
- increasing virtual memory

Student B: You are a support center employee. Answer Student A's questions.

Writing

9 **Use the conversation from Task 8 to complete the receipt.**

ClearPic
Support Center Receipt

Customer
Issue: _____

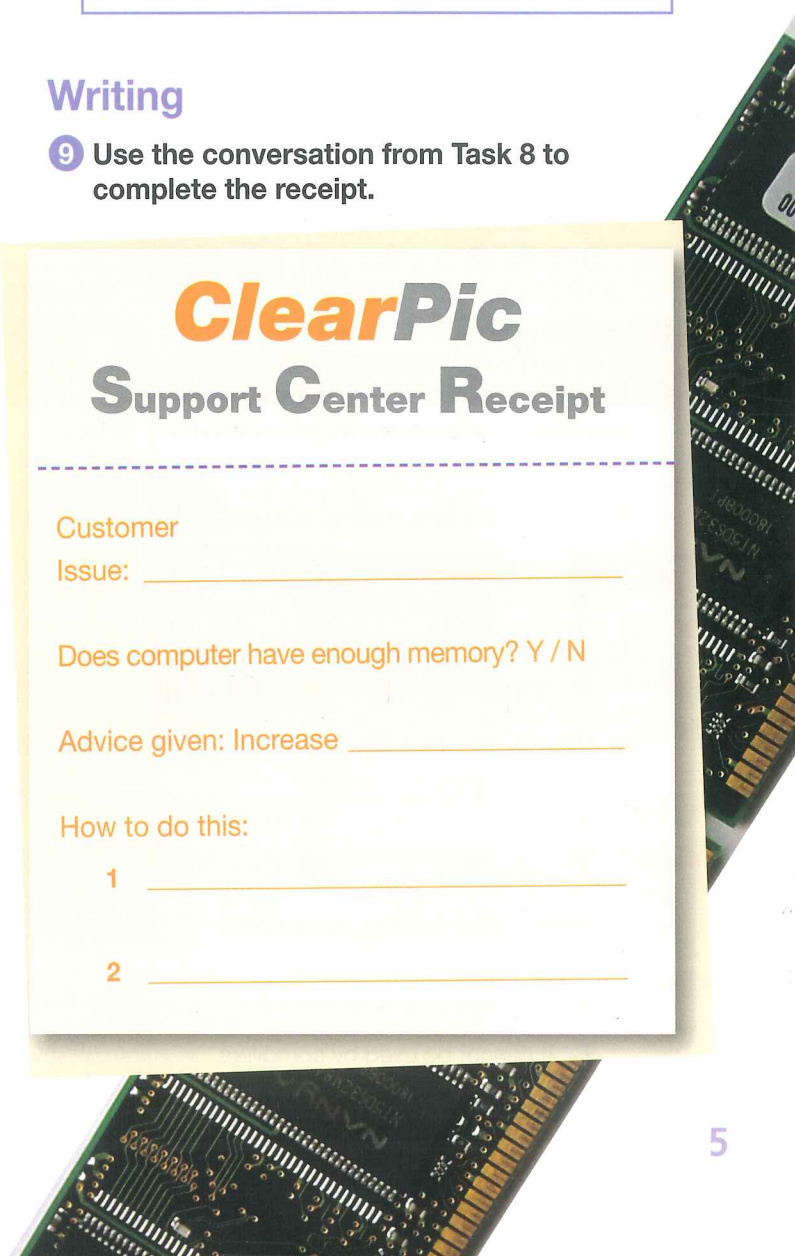
Does computer have enough memory? Y / N

Advice given: Increase _____

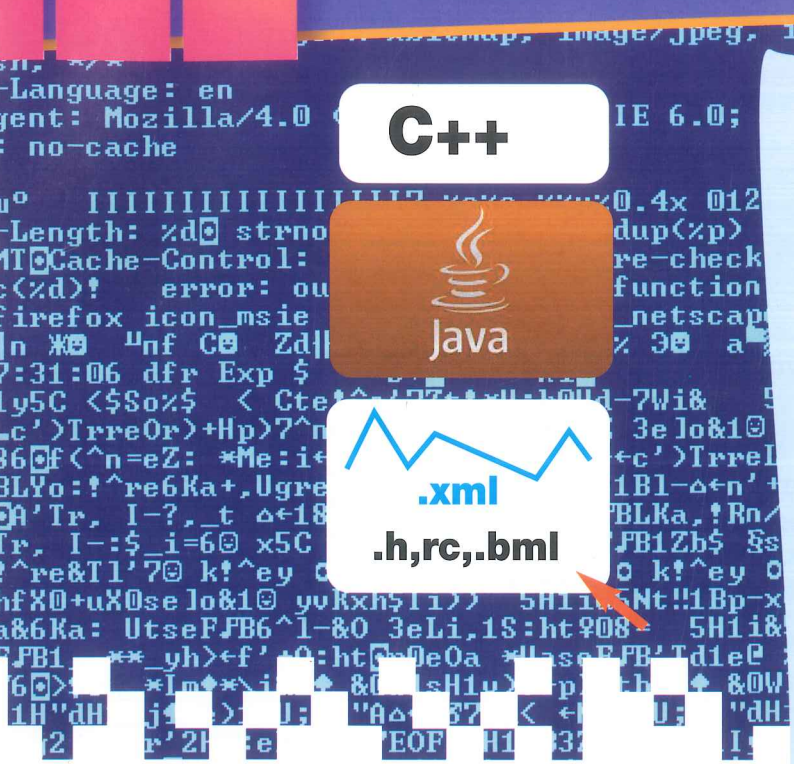
How to do this:

1 _____

2 _____



2 Programming Languages



Mr. Daniels,

As the head of the computer programming department, I feel that we need to update the department's resources. This includes our programmers. Most of our programmers are comfortable with the **C**, **C++**, and **Java computer languages**. We even have one programmer who can use **C#**. But we don't have anyone proficient in **Assembler**. I think we need some programmers who are familiar with it. That way, we can translate between programs more easily.

In addition, we have two programmers who specialize in **BASIC**. This language is in decline and will probably be phased out soon.

Some of our machines are getting older, too. A few computers need their **compilers** and **linkers** updated. Otherwise, they'll never be able to convert to **machine code** efficiently. One of our older **CPUs** cannot **multi-thread** very well. As you know, the programming department is very important to this company's future. Please help us stay updated.

Thanks,
Sarah Jennings

Get ready!

1 Before you read the passage, talk about these questions.

1 How does computer programming affect business?

2 Why is computer programming becoming more important?

Reading

2 Read the letter from the head of a computer programming department to her boss. Then, choose the correct answers.

1 What is the letter mainly about?

A purchasing more CPUs
B writing better computer programs
C learning new computer languages
D requesting updated resources

2 Which of these programming languages do the programmers NOT know?

A C#
B Assembler
C BASIC
D C++

3 What can be inferred about BASIC?

A It is used in Assembler.
B It is used less frequently than C++.
C It cannot be used by compilers.
D It is the most difficult to learn.

Vocabulary

3 Write a word that is similar in meaning to the underlined part.

1 He's writing a new <u>language that computers can process</u>.
c _ _ _ _ _ _ r _ _ _ _ u _ _ _

2 <u>An object-oriented programming language</u> is used often.
_ _ v _

3 I'd use <u>a programming language that translates between programs</u>. _ s s _ _ _ _ _ _

4 Many of the first PCs used <u>an early programming language that is not widely used anymore</u>.
_ _ S _ C

5 Harold often used <u>a competitor to Java</u> instead of Java. _ _

6 Do you know <u>a programming language that adds capabilities to C</u>? _ _+

7 Linux® is created with <u>a programming language that programs operating systems</u>. __

8 The <u>central processing unit</u> is a computer's most important part. _ P _

9 A <u>computer part that changes source code into machine code</u> is essential to programming.
_ _ _ _ _ _ e r

4 **Check (✓) the sentence that uses the underlined parts correctly.**

1 ___ **A** Linkers allow programmers to control computers.
___ **B** CPU was the first programming language.

2 ___ **A** A new Java can make your computer run faster.
___ **B** Machine code makes programming possible.

3 ___ **A** Added memory allows computers to multi-thread.
___ **B** I'm learning compiler, a new programming language.

5 **Listen and read the letter from the head of a computer programming department to her boss. Why will a few of the old computers need their linkers and compilers updated?**

Listening

6 **Listen to a conversation between a programmer and her manager. Mark the following statements as true (T) or false (F).**

1 ___ The manager intends to hire more programmers.
2 ___ The woman is going to learn Assembler.
3 ___ The manager denied the request to upgrade the CPUs.

7 **Listen again and complete the conversation.**

Programmer: Well, without someone who knows Assembler we can't efficiently **1** _____ _____ _____.

Manager: I know, but I think the best way to address that problem is to have you trained in it.

Programmer: I'm up for that. But I'm worried that I'll get behind on the **2** _____ programs I'm writing.

Manager: We can shift some of your workload to other **3** _____, then.

Programmer: Okay. What do you think about our **4** _____ _____?

Manager: How pressing would you say those needs are?

Programmer: If we had some updated CPUs, we could finish our projects in a **5** _____ _____ _____ _____.

Manager: All right. Let me **6** _____ _____ _____ at the budget and see what I can do.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

I don't think we can afford more programmers.

What do you think about our CPU needs?

How pressing would you say those needs are?

Student A: You are talking to a programmer. Ask Student B about:

- the department's needs
- the importance of needs
- the possible solutions

Student B: You are a programming employee. Answer Student A's questions.

Writing

9 **Use the conversation from Task 8 to write an email to your manager to ask for updated resources. Create names for the employee and his/her manager.**

To: _____
From: _____
Subject: _____

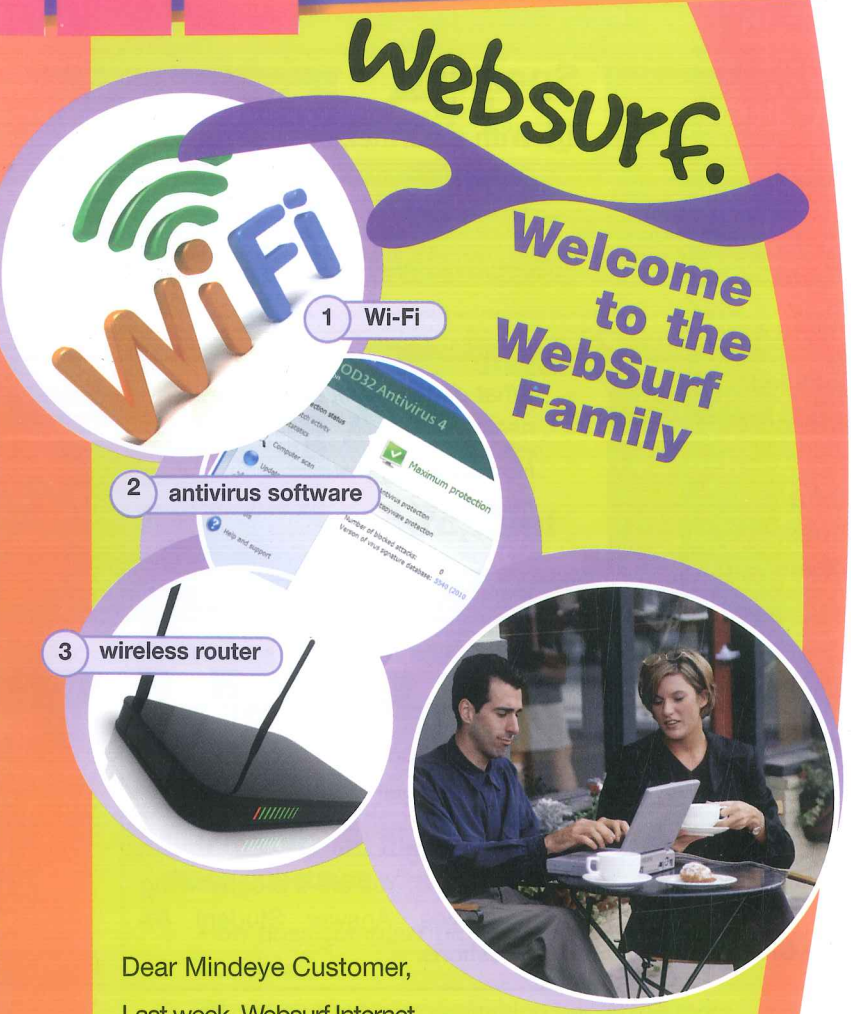
Dear _____

_____
_____
_____

Thanks,

_____

3 ISPs and Internet Access



Dear Mindeye Customer,

Last week, Websurf Internet purchased Mindeye Communications. Starting next month we will be your **ISP**.

This new relationship has many benefits. First, you receive a free download of our exclusive VirusWatch 9.0 **antivirus software** to protect your computer. We're also building all-new **T-3** lines to accommodate the new customers. This means greater bandwidth for **broadband** and **T-1** users.

Unfortunately, we're discontinuing **dial-up** service. But we are **upgrading** to **DSL**. Turn in your old **modem** to us to receive a new DSL modem free of charge.

You also receive our award-winning technical support. We'll help you set up your **wireless router** to bring **Wi-Fi** to your household. We can even help recover your password if you forget.

You'll also have access to WebSurf **hotspots**. You can find them at selected airports, restaurants and coffee shops. All you need is a valid WebSurf account.

We look forward to providing you with the world's best Internet service.

Sincerely,
Gina Matson
Customer Relations,
Websurf Internet

Get ready!

1 Before you read the passage, talk about these questions.

1 How do people access the Internet?
2 What type of Internet connection is best for businesses?

Reading

2 Read the notification from an Internet service provider. Then, choose the correct answers.

1 What is this notice about?
- **A** Mindeye's plan to buy WebSurf
- **B** what WebSurf offers to new customers
- **C** WebSpot offering free wireless routers
- **D** troubleshooting Internet access issues

2 Which service does WebSurf NOT offer?
- **A** tech support
- **B** password selection
- **C** hotspot access
- **D** broadband Internet access

3 What can you infer from the notice?
- **A** Anyone can use WebSurf hotspots.
- **B** More people use T-1 than broadband.
- **C** WebSurf expects to retain Mindeye customers.
- **D** Mindeye offers its customers more services.

Vocabulary

3 Match the words (1-5) with the definitions (A-E).

1 __ T-1
2 __ Wi-Fi
3 __ ISP
4 __ wireless router
5 __ broadband

- **A** an Internet connection using wireless signals
- **B** an Internet connection using regular phone lines
- **C** a device that sends information over phone lines
- **D** lines that allow many users to connect to the Internet
- **E** a system allowing the transfer of large amounts of data

4 **Fill in the blanks with the correct words and phrases from the word bank.**

WORD BANK

antivirus software hotspot ISP upgrade
wireless router DSL T-3

1 This coffee shop has a(n) _____ so customers can go online.
2 A _____ connection only requires a phone line to work.
3 Contact your _____ to get in touch with tech support.
4 _____ my service because I need more bandwidth.
5 Using _____ protects your computer.
6 You need a(n) _____ to get a Wi-Fi network.
7 The new _____ lines offer greater bandwidth

5 **Listen and read the notification from an Internet Service provider again. What service will Websurf Internet put an end to?**

Listening

6 **Listen to a conversation between a customer and a WebSurf customer representative. Mark the following statements as true (T) or false (F).**

1 __ The customer plans to cancel his service.
2 __ The new T-3 lines will increase bandwidth.
3 __ Customers must pay extra for services like tech support.

7 **Listen again and complete the conversation.**

Customer: Just **1** _____ Internet.
Representative: Oh, that's it then. We're in the process of installing new **2** _____ lines.
Customer: But how does that affect my billing?
Representative: T-3 lines are expensive, so we've increased rates. But it allows for greater **3** _____.
Customer: I'm not sure that justifies the price hike.
Representative: Well, it also supports other WebSurf benefits. Live **4** _____ _____, for instance.
Customer: I guess I have used that a few times already.
Representative: You also get free **5** _____ _____. And you can log into any of our **6** _____ at no additional charge.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

What service do you have?
It allows for greater bandwidth.
You can log into hotspots.

Student A: You are calling customer service. Ask Student A about:

- a bill increase
- causes
- benefits

Student B: You work in customer service. Answer Student B's questions.

Writing

9 **Use the conversation from Task 8 to fill in the customer survey.**

Websurf.

Dear Customer,

Thank you for calling WebSurf customer service! In order to help improve the quality of our service, we'd like to ask you a few questions.

What was your call concerning?

_____
_____
_____

Was our Customer Rep able to help you with your problem? How?

_____
_____
_____
_____

4 Storage Devices

Get ready!

1 Before you read the passage, talk about these questions.

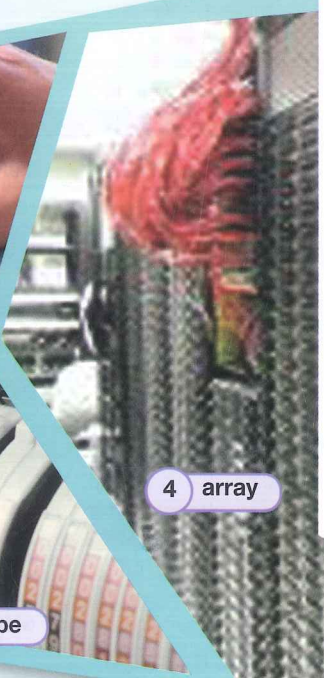
1 Which types of storage devices do people use?

2 What do people use USB flash drives for?

1 USB flash drive

2 CD-R

3 magnetic tape



4 array

Suggestion Form

GENE Tech Systems

Name: Harold Gibbons
Department: IT **Suggestion:**

The IT department is in desperate need of new storage devices. To begin, we need a group of hard drives arranged in an **array.** That means we'll also need **RAID** software. Just last week, an engineer lost months of data in a drive **failure**. An array will help us avoid such costly losses in the future.

Furthermore, many employees are still using outdated devices like **CD-Rs and CD-RWs** to transfer files. Providing each engineer with at least a **USB flash drive** would be much more efficient. Some employees do use their own drives. But state-of-the-art storage devices should be provided by the company.

Finally, GenTech Systems would do well to upgrade the main data backup. We are still using **magnetic tape**. Even using **data compression**, the **memory storage density** of magnetic tape can't compare to modern technology. A 2TB **SATA** drive could store more data at a fraction of the cost.

Reading

2 Read the suggestion form sent from an employee in the IT department to his manager. Then, choose the correct answers.

1 What is the main idea of the form?

A a request for new equipment
B suggestions on improving efficiency
C where to place new storage devices
D how to speed up file transfers

2 Which drive stores more data than magnetic tape?

A CD-R
B RAID
C SATA
D USB flash drive

3 It can be inferred the USB flash drives are...

A less reliable than magnetic tape.
B more efficient than CD-Rs.
C more expensive than a hard drive.
D less efficient than SATA drives.

Vocabulary

3 Fill in the blanks with the correct words and phrases from the word bank.

word BANK

array memory storage density CD-R
data compression CD-RW RAID

1 Set up a(n) _____ to combine drives.

2 The old drive has low _____ and holds little data.

3 A(n) _____ can only be written on once.

4 Without _____ , no more data will fit on that drive.

5 A(n) _____ can be written again and again.

6 _____ software improves hard drive performance.

4 **Check (✓) the sentence that uses the underlined parts correctly.**

1 __ **A** Save the file on the USB flash drive.
__ **B** Arrays are made from a series of floppy disks.

2 __ **A** SATA drives are used to increase processing speeds.
__ **B** A drive failure means the data disk drive is inaccessible.

3 __ **A** RAID arrays are made of magnetic tape.
__ **B** CD-Rs are an older technology.

5 **Listen and read the suggestion form sent from an employee in the IT department to his manager again. What do GENETech Systems use to back up their main data?**

Listening

6 **Listen to a conversation between an IT employee and his manager. Mark the following statements as true (T) or false (F).**

1 __ The company will purchase an array.
2 __ All of the requests were approved.
3 __ Management thinks a new main data backup device is too expensive.

7 **Listen again and complete the conversation.**

Manager: Management didn't like hearing about the **1** _____ _____, so we're getting the array.
Employee: Yes, that was pretty bad. What about some of the **2** _____ _____?
Manager: They weren't approved. It just seemed too expensive.
Employee: **3** _____ _____, but I guess you've got to save money somehow. How about the main data backup?
Manager: We're interested in the savings you mentioned. Could you **4** _____ _____ _____ a more **5** _____ _____?
Employee: Absolutely. When would you like that?
Manager: **6** _____ _____ _____ _____.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

A few of the requests you sent in ....
What about some of the...?
What about the main data backup?

Student A: You are talking to your manager. Ask Student B about:

- approved requests
- unapproved requests
- writing a proposal

Student B: You are the manager. Answer Student A's questions. Make up a name for your subordinate.

Writing

9 **You are an employee in IT. Use the conversation from Task 8 to write a memo to your co-workers telling them which types of new equipment were approved for purchase and why.**

GENE Tech Systems

Memorandum

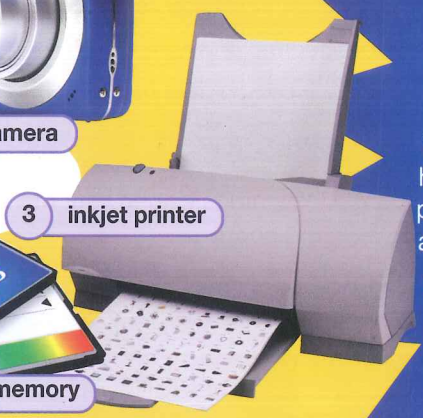
From: _____

To: _____

5 Peripherals

1 automatic document feeder

2 digital camera



3 inkjet printer

4 flash memory

Compumart Blowout Sale Weekend!

This weekend Compumart is cutting prices on hardware to make room for new models.

We have **digital cameras** for both the casual shooter and the professional. With **CCDs,** our cameras take pictures up to 14 **megapixels**, 30x **optical zoom** and 5x **digital zoom**. All sale cameras feature **flash memory**—you can connect to any computer to print out your best snapshots.

We have low prices on both **inkjet printers** and **laser printers**. Our high-end models have **calibration** settings for professional-grade photo prints. Some of our models are all-in-one packages that include scanning and fax.

We're also selling stand-alone scanners. Scan photos and documents at resolutions of up to 4800 **DPI.** Don't forget a **transparency adapter** for scanning slides. Scanning multiple documents? Pick up an **automatic document feeder** and let the machine do the work for you.

Compumart is staying open late all weekend, just for you. Arrive early to get the hardware you need.

5 laser printer

6 transparency adapter

Get ready!

1 Before you read the passage, talk about these questions.

1 How has digital technology changed photography and printing?

2 What types of businesses use scanners, printers and digital cameras?

Reading

2 Read the computer store advertisement. Then, choose the correct answers.

1 What is the advertisement about?

A a new advancement in printer technology
B reduced prices on older hardware models
C Compumart's "going out of business" sale
D accessories for cameras and printers

2 What is NOT true about Compumart's sale?

A They plan to stock up on new models.
B They are cutting prices on software.
C They sell accessories for their scanners.
D They have changed their hours for the sale.

3 What can you infer from the advertisement?

A All new models are on sale.
B Not every digital camera allows flash memory.
C Compumart has scanners that come with auto feeders.
D Some printers cannot create professional grade prints.

Vocabulary

3 Match the words (1-9) with the definitions (A-I).

1 __ DPI
2 __ transparency adapter
3 __ calibration
4 __ automatic document feeder
5 __ inkjet printer
6 __ digital camera
7 __ flash memory
8 __ megapixel
9 __ laser printer

A a device that scans see-through objects
B a device that uses a beam of light to create an image
C a device that uses ink to create an image
D the number of dots used to create an image
E a device used to store pictures on cameras
F a measurement of image size
G the adjustment of devices to correctly display colors
H a device that records images for the computer
I a device that loads papers to be scanned

4 **Fill in the blanks with the correct words from the word bank.**

WORD BANK

optical zoom CCD digital zoom

1 Using _____ blurs a photo's detail.
2 A thicker lens will increase your camera's _____.
3 Digital cameras need a(n) _____ to capture an image.

5 **Listen and read the computer store advertisement again. What do the all-in-one packages for printers incorporate?**

Listening

6 **Listen to a conversation between a salesman and a photographer. Choose the correct answers.**

1 What does the woman want to do?
- **A** return her inkjet printer
- **B** exchange her SuperShot 350
- **C** buy more professional equipment
- **D** research inkjet and a laser printers

2 What can you infer about the woman?
- **A** Her last inkjet printer broke.
- **B** She is buying her first digital camera.
- **C** She plans on selling her old equipment.
- **D** She wants to improve the quality of her work.

7 **Listen again and complete the conversation.**

S: May I recommend the SuperShot 350? It has 15 times **1** _____ _____ and two times digital zoom.
P: That's pretty much what I have now. I'm looking for something a bit **2** _____ _____.
S: Okay ... how about the SnapPro series? This one has 30 times optical zoom and 5 times digital.
P: How many **3** _____ does it have?
S: Fifteen, which gives you incredible detail.
P: That **4** _____ _____. Now, I'd also like to start printing my own pictures.
S: Well, we have some great deals on inkjet and **5** _____ _____.
P: I'm leaning towards a laser printer. Inkjets just can't print as well.
S: Okay, the PrintBeam EX is a two-in-one laser printer and scanner. It prints at 2400 **6** _____.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

May I recommend the ...?
This one has ...
I'd also like to start ...

Student A: You are salesman. Talk to Student B about:
- what he or she needs
- desired specifications

Student B: You are a customer. Choose a product you need and answer Student A's questions.

Writing

9 **Use the conversation from Task 8 to fill in the customer's review on a discussion board. Talk about what item you bought, the specifications of the item, and what you will use it for.**

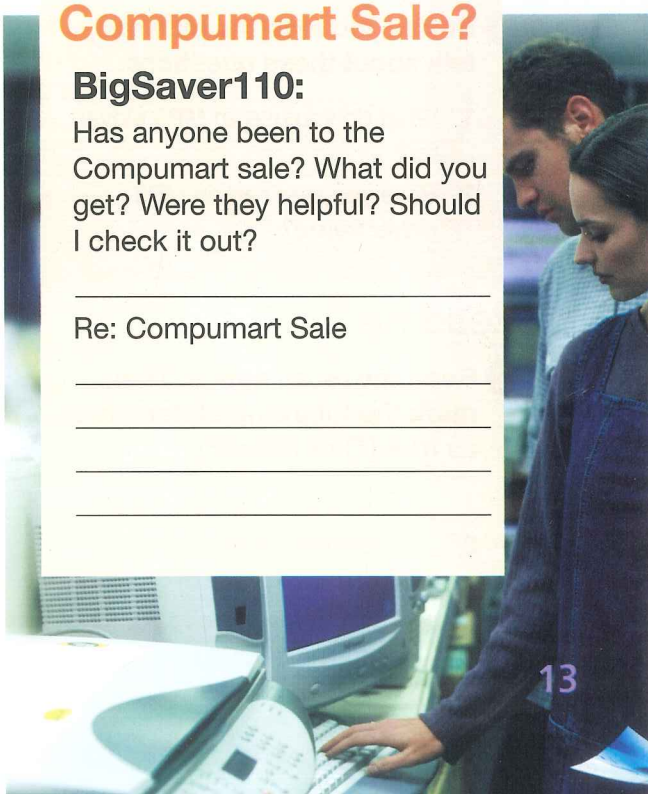
Compumart Sale?

BigSaver110:
Has anyone been to the Compumart sale? What did you get? Were they helpful? Should I check it out?

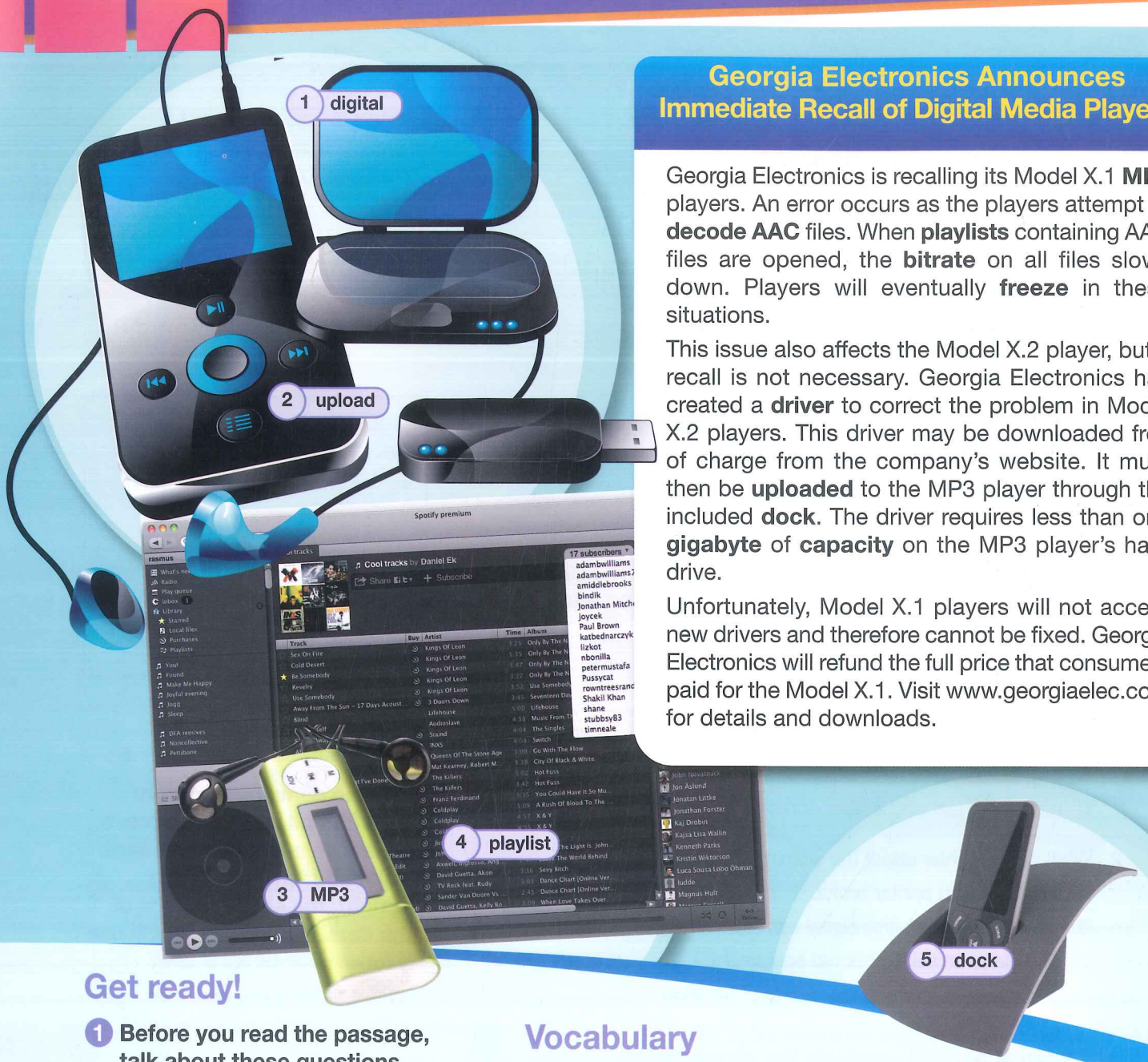
_____

Re: Compumart Sale

_____
_____
_____
_____



6 MP3 Players



Georgia Electronics Announces Immediate Recall of Digital Media Players

Georgia Electronics is recalling its Model X.1 **MP3** players. An error occurs as the players attempt to **decode AAC** files. When **playlists** containing AAC files are opened, the **bitrate** on all files slows down. Players will eventually **freeze** in these situations.

This issue also affects the Model X.2 player, but a recall is not necessary. Georgia Electronics has created a **driver** to correct the problem in Model X.2 players. This driver may be downloaded free of charge from the company's website. It must then be **uploaded** to the MP3 player through the included **dock**. The driver requires less than one **gigabyte** of **capacity** on the MP3 player's hard drive.

Unfortunately, Model X.1 players will not accept new drivers and therefore cannot be fixed. Georgia Electronics will refund the full price that consumers paid for the Model X.1. Visit www.georgiaelec.com for details and downloads.

Get ready!

1 Before you read the passage, talk about these questions.

1 What do you use an MP3 player for?

2 Have you ever had an MP3 player freeze?

Reading

2 Read the recall notice. Then, mark the following statements as true (T) or false (F).

1 __ The recall is for Model X.1.

2 __ The driver will take up less than one gigabyte of capacity.

3 __ The issue is related to the decoding of MP3 files.

Vocabulary

3 Match the words (1-9) with the definitions (A-I).

1 __ digital	4 __ gigabyte	7 __ playlist
2 __ AAC	5 __ decode	8 __ dock
3 __ capacity	6 __ MP3	9 __ bitrate

A a file format that is higher quality than MP3s

B a list of songs chosen by a user

C a device for connecting an MP3 player to a computer

D the amount of room available on a storage device

E a unit of memory

F the standard format for encoding sound files

G a measurement of the amount of data processed per second

H to change a file back to its original form

I storing information in a series of ones and zeros

4 **Fill in the blanks with the correct words from the word bank.**

WORD BANK

upload freeze driver

1 Did you _____ the file yet?

2 Computers _____ when too many programs are open.

3 You must install a(n) _____ with all new hardware.

5 **Listen and read the recall notice again. Why isn't the Model X.2 player being recalled?**

Listening

6 **Listen to a conversation between a consumer and a company representative. Choose the correct answers.**

1 What are the speakers mostly talking about?

A why AAC files are causing problems
B where a consumer can buy a driver
C how to solve a consumer's problem
D which model a consumer owns

2 How did the customer try to fix the MP3 player?

A install a new driver
B restart the player
C upload a new playlist
D remove the AAC files

7 **Listen again and complete the conversation.**

C: Well, I've **1** _____ _____ _____ with my MP3 player.

R: Yes, we're **2** _____ _____ _____ on some models. Which model do you have?

C: I have the X.2.

R: All right. Tell me about the problem you're having.

C: At first, the sound quality started getting worse. Then it **3** _____ _____ on me. I thought restarting it would fix it, but nothing works.

R: Okay, that's a problem caused by **4** _____ files. Luckily, we have a driver to fix that.

C: Oh good. How do I get it?

R: Just download it from our website and **5** _____ it to your player.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

I've got a problem with my ...
Which model do you have?
Tell me about the problem you're having.

Student A: You are a company representative. Talk to Student B about:

- a recall
- one model
- problems

Student B: You are calling to get help with your MP3 player. Answer Student A's questions.

Writing

9 **You are a customer experiencing problems with your MP3 player. Use the conversation from Task 8 to write a letter to the company telling them about your problems. Make up a name for the customer.**

Dear Sir or Madam,

_____

_____

_____

_____

_____

_____

_____

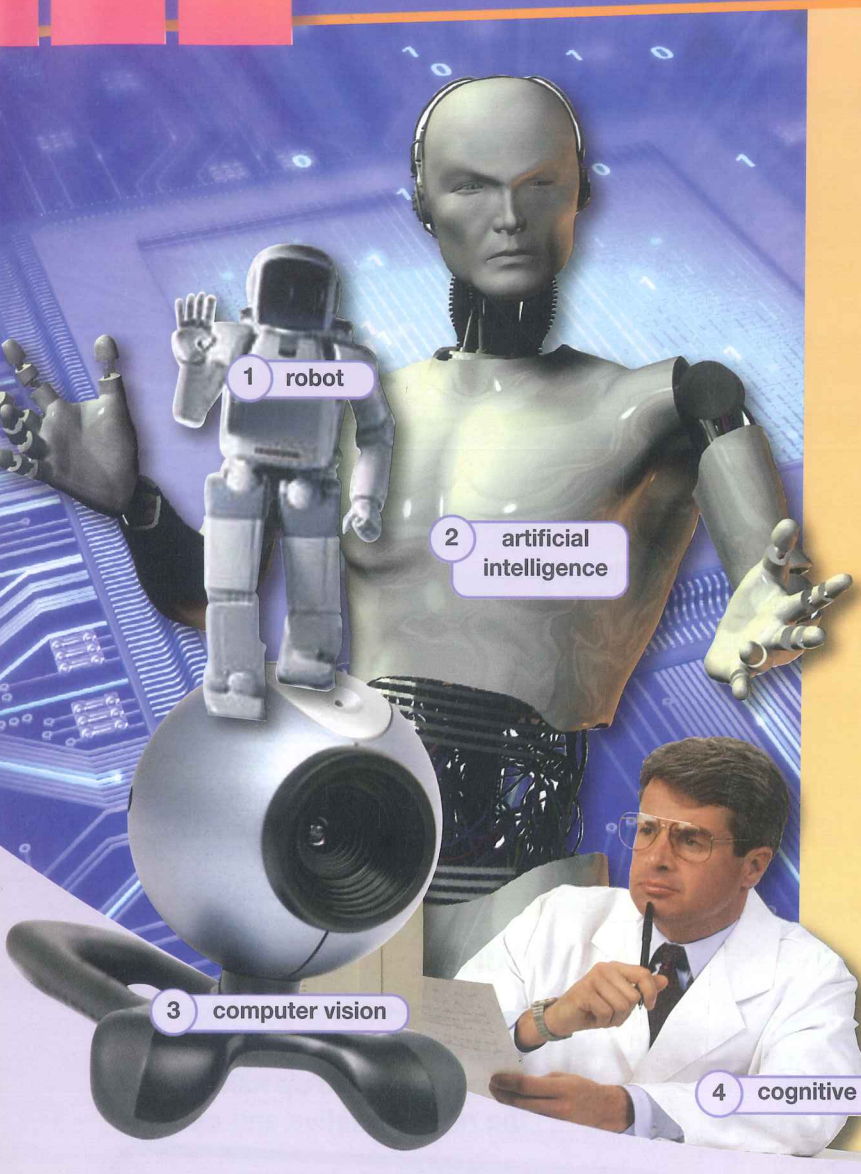
_____

Sincerely,

_____

_____

15 Artificial Intelligence



The Monthly Programmer | January Edition

Artificial Intelligence, the Future of Technology

Will **Heuristics** Lead to **Consciousness**?

By Peter Smith

Computers perform complex calculations that no human could complete. Yet, they remain unable to think for themselves. That's something **cognitive scientists** want to change. Their research focuses on two types of thinking: **case based** and **model based reasoning**. Case based reasoning compares similar problems. Model based reasoning examines cause and effect. But these different approaches have the same problem. Their **algorithms**, like most rules, aren't always right.

That's where heuristics comes in. With heuristics, computers are avoiding that problem. Instead of following rigid guidelines like regular algorithms, heuristics search for acceptable solutions. As these modified algorithms improve, **machine learning** leaps forward. Advanced computer **agents** and **robots** now find solutions independently. **Pattern recognition** has even made **computer vision** possible. The range of what robots and computers can do grows each day. The invention of a self learning computer may soon be possible.

Cause → Effect

5 model based reasoning

Get ready!

1 Before you read the passage, talk about these questions.

1 Do you think robots will ever think like humans?
2 Will artificial intelligence help or hurt business? Why?

Reading

2 Read the magazine article on Artificial Intelligence. Then, mark the following statements as true (T) or false (F).

1 __ Case and model based reasoning face different complications.
2 __ Heuristics don't have rigid guidelines like regular algorithms.
3 __ Computer vision is improving pattern recognition.

Vocabulary

3 Match the words (1-6) with the definitions (A-F).

1 __ pattern recognition
2 __ cognitive scientist
3 __ consciousness
4 __ robot
5 __ heuristic
6 __ algorithm

A a machine capable of independent action
B a set of rules to solve a problem that seeks any solution
C the process of classifying objects by appearance
D a sense of oneself and one's surroundings
E a rule or set of rules to solve a problem with limited solutions
F a person who studies how the mind works

4 **Write a word that is similar in meaning to the underlined part.**

1 Giving machines the ability to see is improving.
c _ _ _ u _ _ _ _ i _ _ o _

2 Examining previous experiences to make decisions works best.
m _ _ e l b _ _ _ _ _ e _ _ o _ _ n g

3 I use examinations of cause and effect relationships to make decisions.
_ a _ e _ a _ _ _ r _ _ s _ n _ _ _ _

4 Peter works in the science that attempts to create thinking robots.
_ r _ _ f _ _ i _ _ i _ _ e _ _ i _ _ n _ _

5 This computer that reacts to an environment turns the heat on when it's cold.
_ g _ _ t

6 The field of creating algorithms that computers can learn from often uses heuristics.
m _ _ _ i _ _ _ _ _ r _ i _ g

5 **Listen and read the magazine article on Artificial Intelligence again. What has made computer vision feasible?**

Listening

6 **Listen to a conversation between two programmers. Mark the statements as true (T) or false (F).**

1 __ The woman published an article.
2 __ The man believes that cognitive scientists will harm programming.
3 __ An algorithm needs a perfect solution.

7 **Listen again and complete the conversation.**

Programmer 1: Jane, did you **1** _____ _____ _____ to read this month's Programmer?

Programmer 2: Yes, I saw your article on AI. So **2** _____ _____ are getting into programming?

Programmer 1: Yes, and I think it'll help. Most programming is too rigid.

Programmer 2: Because of algorithms? I **3** _____ _____ that part.

Programmer 1: Basically, an algorithm needs a **4** _____ answer.

Programmer 2: **5** _____ _____ _____ _____ by, "needs a perfect answer"?

Programmer 1: They have a limited number of solutions. If one of those solutions isn't right, the algorithm doesn't work like a math problem.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

Yes, I saw your article on AI.

I didn't get that part.

They have a limited number of solutions.

Student A: You are a programmer. Ask Student B about:

- cognitive scientists
- algorithms

Make up a name for your colleague.

Student B: You are a programmer. Answer Student A's questions.

Writing

9 **ICT** **Collect information about Artificial Intelligence (A.I.). Then, write a paragraph explaining what it is. You can visit this website:** ***http://www.alanturing.net/turing_archive.***

Glossary

4G network [N-COUNT-U7] A **4G network** is a wireless network that sends and receives data very quickly.

AAC [N-UNCOUNT-U6] **AAC** is a music file format that is of higher quality than MP3.

account activity [N-UNCOUNT-U11] **Account activity** is a withdrawal, deposit or transfer in a bank account.

account alert [N-COUNT-U11] An **account alert** is a service that notifies bank customers of suspicious activity.

accuracy [N-COUNT-U10] **Accuracy** is how correct a GPS's measurement is.

acquisition time [N-UNCOUNT-U10] **Acquisition time** is how long a GPS takes to determine location.

active matrix display [N-COUNT-U14] An **active matrix display** is a type of screen that produces high quality, clear images.

actuator [N-COUNT-U13] An **actuator** is a piece of equipment that uses energy to make a robot move.

agent [N-COUNT-U15] An **agent** is a computer that observes and reacts to the conditions in a certain environment.

algorithm [N-COUNT-U15] An **algorithm** is a rule or set of rules that provides steps to take or questions to ask in order to solve a problem.

antivirus software [N-UNCOUNT-U3] **Antivirus software** is a program that locates and eliminates computer viruses.

app [N-COUNT-U7] An **app** is an application with a specific function that you download onto your phone.

array [N-COUNT-U4] An **array** is a group of disk drives that are connected and used as a single unit.

Artificial Intelligence [N-UNCOUNT-U15] **Artificial Intelligence** is the branch of science that seeks to create computers and robots that can reason, learn and solve problems independently.

Assembler [N-UNCOUNT-U2] **Assembler** is a low level programming language that translates between computer programs.

attachment [N-COUNT-U8] An **attachment** is a file sent as part of an email.

automatic document feeder [N-COUNT-U5] An **automatic document feeder** (ADF) is a device that feeds multiple pages into a scanner.

autonomous [ADJ-U13] A robot that is **autonomous** can operate on its own without help from humans.

avatar [N-COUNT-U9] An **avatar** is an image in a computer game that represents a person.

bank fraud [N-UNCOUNT-U11] **Bank fraud** is a crime in which someone uses bank funds without permission.

barcode [N-COUNT-U8] A **barcode** is an arrangement of lines and spaces that represent data.

BASIC [N-UNCOUNT-U2] **BASIC** is a high level programming language that was invented in the 1960s and is declining in popularity.

bill pay [N-UNCOUNT-U11] **Bill pay** is a service provided by banks that allows customers to automatically pay bills online.

bit [N-COUNT-U1] A **bit** is the smallest unit of computer data.

bit size [N-COUNT-U1] A computer's **bit size** measures the number of bits that the CPU can access at once.

bitmapped graphic [N-COUNT-U14] A **bitmapped graphic** is an image that is made up of many small points.

bitrate [N-UNCOUNT-U6] **Bitrate** is a measurement of the amount of data that is processed.

black and white [ADJ-COUNT-U14] An image or picture that appears only in black, white, and varying shades of gray is **black and white**.

blog [N-COUNT-U9] A **blog** is a website where a person writes posts about his or her feelings, thoughts, activities, etc.

Bluetooth [N-UNCOUNT-U7] **Bluetooth** is a wireless network that connects your phone with other devices, such as a computer.

boolean operator [N-COUNT-U12] A **boolean operator** narrows a database search with words like "and," "or" and "not."

broadband [N-UNCOUNT-U3] **Broadband** is a system that allows computers to transfer large amounts of information.

bus speed [N-COUNT-U1] A computer's **bus speed** measures how many times the CPU can access a group of bits per second. This is measured in MHz.

C [N-UNCOUNT-U15] **C** is a programming language that is used to program operating systems.

C# [N-UNCOUNT-U15] **C#** is a programming language that was invented as an alternative to Java.

C++ [N-UNCOUNT-U15] **C++** is a programming language that adds object-oriented programming capabilities to C.

calibration [N-UNCOUNT-U5] **Calibration** is the act of adjusting printers and scanners to display colors correctly.

call waiting [N-UNCOUNT-U7] **Call waiting** is a feature of some cell phones. It alerts you that someone is calling while you are on the line with someone else.

caller ID [N-UNCOUNT-U7] **Caller ID** is a feature of some cell phones that tells you the name and number of the person calling you.

capacity [N-COUNT-U6] **Capacity** is the amount of space an MP3 player has to store files.

cartography [N-UNCOUNT-U10] **Cartography** is the art of mapmaking.

case based reasoning [N-UNCOUNT-U15] **Case based reasoning** is the process of using information from previous experiences to solve new problems.

catalog [N-COUNT-U12] A **catalog** is an inventory of books in a library.

CCD [N-COUNT-U5] **CCD** (charge-coupled device) is a light-sensitive chip that is used when capturing digital images.

CD-R [N-COUNT-U4] A **CD-R** is a blank CD that can have information written on it only once.

CD-RW [N-COUNT-U4] A **CD-RW** is a blank CD that can have information written on it multiple times.

cell phone [N-COUNT-U7] A **cell phone** is a wireless phone that sends and receives data at a certain frequency of energy.

cognitive scientist [N-COUNT-U15] A **cognitive scientist** is a person who studies the mind and thought processes.

compiler [N-COUNT-U15] A **compiler** changes human-readable source code into machine code that a CPU can understand.

computer language [N-COUNT-U15] A **computer language** (also known as a programming language) is used by computer programmers to create instructions that a computer can understand.

computer vision [N-UNCOUNT-U15] **Computer vision** is the field and technology of making computers that have the ability to see.

consciousness [N-UNCOUNT-U15] **Consciousness** is the state of being aware of oneself and one's actions.

coordinates [N-COUNT-U10] **Coordinates** are groups of numbers that describe location.

CPU [N-UNCOUNT-U15] A **CPU** (Central Processing Unit) is the part of a computer that handles most of the processing of files and data.

data compression [N-UNCOUNT-U4] **Data compression** is the process of storing information using less storage space by reducing the number of bits it takes up.

data plan [N-COUNT-U8] A **data plan** is a contract from a cell phone service provider that states what online services are available on a cell phone and how much that transfer of data will cost.

decode [V-T-U6] To **decode** a file is to return it to its original format.

digital [ADJ-U6] If a file is **digital**, it is stored electronically as a series of ones and zeros.

digital camera [N-COUNT-U5] A **digital camera** is a device that records images to be looked at on a computer.

digital zoom [N-COUNT-U5] **Digital zoom** is how a digital camera increases the display size of an image subject, reducing image quality.

DIMM [N-COUNT-U1] A **DIMM** (Dual Inline Memory Module) is a newer type of memory module. It sends 64 bits of data to the CPU.

dock [N-COUNT-U6] A **dock** is a device that connects an MP3 player to a computer.

download [V-COUNT-U14] To **download** something is to transfer a file or program from the Internet to a personal computer or device.

DPI [N-COUNT-U5] **DPI** (dots per inch) is the number of dots used per inch to create an image.

drive power [N-UNCOUNT-U13] **Drive power** is the energy that makes a robot move.

driver [N-COUNT-6] A **driver** tells a device how it should communicate with a computer.

DSL [N-UNCOUNT-U3] **DSL** (digital subscriber line) is a high-bandwidth connection using normal telephone lines.

dynamic model [N- COUNT-U13] A **dynamic model** is a mathematical model of the forces that allow a robot to move.

Glossary

ebook [N-COUNT-U12] An **ebook** is a book that is available online.

ejournal [N-COUNT-U12] An **ejournal** is a scholarly publication that is published online.

electronic funds transfer [N-COUNT-U11] An **electronic funds transfer** is the act of moving money from one account to another by using a computer.

electronic statement [N-COUNT-U11] An **electronic statement** is a summary of account activity that is viewable on a computer, but not on paper.

elevation [N-COUNT-U10] **Elevation** is the height of something above sea level.

email [N-COUNT-U8] An **email** is an electronic message sent from one user to another.

enabling device [N-COUNT-U13] An **enabling device** is a device that a person activates to allow a robot to move.

end-effector [N-COUNT-U13] An **end-effector** is the hand or tool at the end of a robot's manipulator. It is the part of the robot that moves or picks up objects.

EPC [N-COUNT-U8] An **EPC** (electronic product code) is a number used to identify specific individual items.

equality constraint [N-COUNT-U13] An **equality constraint** is a value that specifies the end-effector's position and movements.

ergonomic [ADJ-COUNT-U14] Something that is **ergonomic** has been designed to prevent injury from using it repeatedly.

EZproxy [N-COUNT-U12] **EZproxy** is a library server program that allows remote users to access the library's electronic materials.

failure [N-COUNT-U4] A **failure** occurs when a disk drive malfunctions and the data stored on it cannot be accessed.

feed [N-COUNT-U9] A **feed** is a format that lets you easily read updated news about something.

flash memory [N-UNCOUNT-U5] **Flash memory** is a data storage medium that is used with digital cameras.

forum [N-COUNT-U9] A **forum** is a website where people can discuss a certain topic.

freeze [V-I-U6] To **freeze** is to become unresponsive and stop functioning.

gaming [N-UNCOUNT-U9] **Gaming** is the activity of playing games online on a computer.

gigabyte [N-COUNT-U6] A **gigabyte** is a unit of measurement of hard drive capacity.

GPS [N-COUNT-U10] **GPS** (Global Positioning System) is a collection of 24 satellites that give accurate position and navigation data to people all over the Earth.

handle [N-COUNT-U9] A **handle** is a name that a person creates to access an account on a computer or website.

hard drive [N-COUNT-U1] A **hard drive** is a device in a computer that stores all of its information permanently.

heuristic [N-COUNT-U15] A **heuristic** is a type of algorithm designed to find a suitable solution to a problem when no perfect solution or answer is known.

hotspot [N-UNCOUNT-U3] A **hotspot** is a location (like an airport or a cafe) that provides wireless Internet access to users.

identity theft [N-UNCOUNT-U11] **Identity theft** is the crime of using other people's personal information to take their money or buy items in their name.

index [N-COUNT-U12] An **index** is a categorical list of books in a library.

inkjet printer [N-COUNT-U5] An **inkjet printer** is a device that uses ink to produce documents and images on paper.

instant message [N-COUNT-U8] An **instant message** is a text-based conversation in real time.

integrity [N-UNCOUNT-U10] **Integrity** is a device's ability to perform consistently without error.

ISP [N-COUNT-U3] An **ISP** (Internet service provider) is a company that provides Internet access.

Java [N-UNCOUNT-U15] **Java** is an object-oriented programming language.

joint [N-COUNT-U13] **Joints** are the places where the parts of a robot connect. They allow the robot's parts to rotate.

laser printer [N-COUNT-U5] A **laser printer** is a device that produces documents and images on paper with a laser beam.

latitude [N-COUNT-U10] **Latitude** is an object's location north or south of the Equator.

license restriction [N-COUNT-U12] A **license restriction** limits who can use a library's electronic resources.

link [N-COUNT-U9] A **link** is a word or set of words on a website or document. When you click on it, it takes you to a different website.

linker [N-COUNT-U15] A **linker** takes the machine code created by a compiler and makes it into an executable file.

lithium-ion battery [N-COUNT-U7] A **lithium-ion battery** is battery in cell phones that is small, light, and lasts a long time.

lockout [N-COUNT-U11] A **lockout** prevents people from accessing a bank account when the password is entered incorrectly too many times.

longitude [N-COUNT-U10] **Longitude** is an object's location east or west of the Prime Meridian.

machine code [N-UNCOUNT-U2] **Machine code** is made up of the instructions that a CPU can process.

machine learning [N-UNCOUNT-U15] **Machine learning** is a field of study concerned with creating algorithms that computers can use to learn from their experiences.

magnetic tape [N-UNCOUNT-U4] **Magnetic tape** is a long, narrow piece of magnetized plastic that is used for storing information.

manipulator [N-COUNT-U13] The **manipulator** is the arm of a robot. The end-effector sits at the end of it.

MB [N-COUNT-U1] A **MB** (megabyte) is a unit of measurement that measures how much information a computer stores.

media [N-UNCOUNT-U12] **Media** are the forms of mass communication available at the library, including images, films and periodicals.

megapixel [N-COUNT-U5] A **megapixel** is a measurement of image size.

memory storage density [ADJ-U4] **Memory storage density** is the amount of data that can be contained within a given computer storage device.

MHz [N-COUNT-U1] A **MHz** (megahertz) is a unit of measurement that measures the speed of a CPU.

MMORPG [N-COUNT-U9] A **MMORPG** (Massive Multiplayer Online Role-Playing Game) is a type of online computer game in which many people interact with each other in a virtual world.

model based reasoning [N-UNCOUNT-U15] **Model based reasoning** is the process of making conclusions by examining cause and effect relationships.

motherboard [N-COUNT-U1] A **motherboard** is the circuit board in a computer. It holds the CPU and all attachments.

MP3 [N-UNCOUNT-U6] **MP3** is the standard format for digital audio files.

multi-thread [V-I-U2] To **multi-thread** is to use a computer to execute several threads of execution simultaneously.

navigation [N-UNCOUNT-U10] **Navigation** is the act of creating a route to a destination.

offline [ADJ-U14] If something is available **offline**, it can be used when an electronic device is not connected to the Internet.

online banking [N-UNCOUNT-U11] **Online banking** is a service that allows people to manage finances on the Internet.

optical zoom [N-COUNT-U5] **Optical zoom** is the use of the camera lens to make an image appear larger, retaining image quality.

paging file [N-COUNT-U1] A **paging file** is the area on the hard drive where the computer stores data that doesn't fit in the RAM.

pattern recognition [N-UNCOUNT-U15] **Pattern recognition** is the process of classifying objects by examining their visual appearance.

PDF reader [N-COUNT-U14] A **PDF reader** is a program that can open and display text or images.

pendant [N-COUNT-U13] A **pendant** is a device that a person uses to control and guide a robot's movements.

PIN [N-COUNT-U11] A **PIN** (Personal Identification Number) is a number assigned to bank account holders that helps banks identify the holder and maintain security.

playlist [N-COUNT-U6] A **playlist** is a list of songs created by an MP3 player user.

position [N-COUNT-U10] A **position** is an object's exact location.

post [N-COUNT-U9] A **post** is a message or news item on a website.

Glossary

profile [N-COUNT-U9] A **profile** is a webpage on a social networking site with personal or identifying information.

publications [N-COUNT-U14] A **publication** is book or magazine that a publishing company releases to the public.

RAID [N-UNCOUNT-U4] **RAID** is a method for combining multiple disk drives into a single storage unit that can increase computer performance dramatically.

RAM [N-UNCOUNT-U1] **RAM** (Random Access Memory) is a form of computer data storage. It is a physical device that holds data temporarily.

reflow [N-COUNT-U14] **Reflow** is a process that allows images or text to fit screens that are different sizes.

remote access [N-UNCOUNT-U12] **Remote access** is the ability to use online library resources from outside the library.

ringtone [N-COUNT-U7] A **ringtone** is a sound or song that a cell phone plays when someone calls it.

robot [N-COUNT-U15] A **robot** is a moving machine that can complete tasks automatically.

robotics [N-UNCOUNT-U13] **Robotics** is the scientific study of robots and their structure and movement.

SATA [N-UNCOUNT-U4] **SATA** is an interface for connecting storage devices to a computer system.

scan [V-T-U8] To **scan** a product is to use a laser to read bar codes and convert them to electronic signals.

serial robot [N-COUNT-U13] A **serial robot** is a robot that consists of a single series of joints linked together.

SIMM [N-COUNT-U1] A **SIMM** (Single Inline Memory Module) is an older type of memory module. It sends up to 32 bits of data to a CPU.

Site key [N-COUNT-U11] A **site key** is a security system that lets the bank and the account holder verify their identities to one another in online transactions.

site license [N-COUNT-U12] A **site license** allows users to access software regardless of their location.

smartphone [N-COUNT-U8] A **smartphone** is a cellular phone that also functions like a PDA.

social networking [N-UNCOUNT-U9] **Social networking** is the activity of socializing online with other people. This takes place on social networking sites, forums, blogs, etc.

spam [N-UNCOUNT-U8] **Spam** is unwanted email, like advertisements, sent to large groups of people. Some email providers direct spam to special folders so that users do not have to see or open it.

SSL connection [N-COUNT-U11] An **SSL** (Secure Sockets Layer) **connection** is a connection between two servers that encrypts information so that no one else can monitor or access the information exchanged.

sub-pixel rendering [N-UNCOUNT-U14] **Sub-pixel rendering** is a process that increases the clarity of text on certain displays.

subscription [N-COUNT-U12] A **subscription** is the agreement to use something, often requiring a fee.

system requirements [N-COUNT-U14] **System requirements** are the hardware or software needed to run a program.

T-1 [N-UNCOUNT-U3] **T-1** is a type of line used by businesses that connects many users at once to the Internet.

T-3 [N-UNCOUNT-U3] **T-3** is a powerful line made of 28 T-1 lines.

tablet [N-COUNT-U8] A **tablet** is a computer that is smaller than a laptop with a touch screen and is used to browse the Internet and check email.

text [V-T-U8] To **text** someone is to send them a text message.

text message [N-COUNT-U7] A **text message** is a short, written message that a person sends to and from a cell phone.

touchscreen [N-COUNT-U7] A **touchscreen** is a display on some cell phones that you tap to access its features and applications.

transparency adapter [N-COUNT-U5] A **transparency adapter** is a device that is used with a scanner to capture see-through documents like slides.

UPC [N-COUNT-U8] A **UPC** (universal product code) is a 12-digit number used to identify consumer items.

upgrade [V-T-U3] To **upgrade** is to improve something, like a computer program or a type of service.

upload [V-T-U6-U14] To **upload** a file is to transfer it from a computer or device to the Internet or to another device, such as an MP3 player.

USB flash drive [N-COUNT-U4] A **USB flash drive** is a portable flash memory system that uses a USB interface to connect to a computer and transfer information.

virtual memory [N-UNCOUNT-U1] **Virtual memory** is a method of computer data storage. The computer stores data on its hard drive instead of in the RAM.

virtual world [N-COUNT-U9] A **virtual world** is an online world where you can meet and interact with other people or computer players.

voice activation [N-UNCOUNT-U7] **Voice activation** is a feature of some cell phones that lets you dial a number by voice instead of by hand.

voicemail [N-UNCOUNT-U7] **Voicemail** is a feature that lets you listen to or leave an electronic spoken message on a cell phone.

VPN [N-COUNT-U12] A **VPN** (virtual private network) is a way to encrypt a connection when you use remote access.

waypoints [N-COUNT-U10] **Waypoints** are locations that you can mark or record in your GPS.

Wi-Fi [N-UNCOUNT-U3] **Wi-Fi** (Wireless Fidelity) is a type of connection using wireless signals instead of a physical connection.

wireless router [N-COUNT-U3] A **wireless router** is a device that forwards data as a wireless signal.

CAREER PATHS

Information Technology

Book 3

Virginia Evans
Jenny Dooley
Stanley Wright

Express Publishing

Scope and Sequence

Unit	Topic	Reading context	Vocabulary	Function
1	Macs and PCs	Magazine article	Apple, color matching, Mac, Microsoft, OS X, PC, prepress, processing speed, typography, video subsystem, Windows	Stating a preference
2	Linux	Email	command lines, free software license, interface, Linux, Linux distribution, Microsoft Office, open source, OpenOffice, TCP/IP protocol, terminal, Ubuntu	Listing concerns
3	Connecting to Clients and Employees	Email	anchor, applet, CAD, CBT, DVD, electronic bulletin board, FAQs, media player, MPEG, multimedia, reliability	Giving an opinion
4	Advertising and Marketing	Contract	animated GIF, click-through rate, click tracking, CPM, CPC, above the fold, commission, campaign, geo targeting, frequency capping, hang	Confirming information
5	Automation	Investor report	automation, CAM, embedded systems, encoder, Ethernet network, fault tolerant, integrated circuits, latency, open system, oscilloscope, priority inversion, redundancy	Asking for suggestions
6	Telecommuting	Letter	Basecamp, centralize, color-coding, gallery, message board, real time, Skype, telecommuting, toggle, tour, web-based	Describing functions
7	Hackers and Viruses	Newspaper article	anti-antivirus virus, hacker, imbed, infect, log in, overwriting virus, piggybacking, replicate, resident extension, resident virus, virus	Discussing possible consequences
8	Identity Theft	Letter	card scanning, cyber, fraud, harming, identity theft, malware, phishing, pretexting, scanner, spyware, Trojan horse, worm	Discussing suspicions
9	Preventative Measures	Email	attack, audit log, authenticated, backdoor, bug, firewall, intruder, keylogger, popup, protocol, security	Checking certainty
10	Anti-virus Software	Advertisement	anti-virus software, backup, block, cyber criminal, detect, false negative, false positive, rootkit, security software, sweep, virus removal software, webpage analysis	Discussing capabilities
11	Cloud Computing	Newspaper column	CDN, cloud computing, deployment, elasticity, PaaS, pay as you go, private cloud, public cloud, SaaS, scalability, utility computing	Expressing a positive opinion
12	Improving Health	Magazine article	ADM, bar coding, CDSS, CPOE, electronic health record, EMM, implanted microchip, interoperability, lab-on-a-chip, micro robotic tweezers, optic scanner, PACS	Giving instructions
13	Creative Design	Website	animated GIF, blog, egress, ingress, interface, keyword, mask, natural language, PCI compliance, redesign, trust logo, web portal	Offering solutions
14	System Administration	Job posting	configure, cover letter, duty, file archiving, install, performance tuning, purge, résumé, system administrator, technical, troubleshooting, Unix	Talking about qualifications
15	Programming	Job listings	back end, editor, GUI component, HTML, interactive, PHP, programmer, scripts, user-friendly, visual basic, web developer, XML	Discussing degree of difficulty

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7 Hackers and Viruses

Wilmington Daily Post

LINCOLNSHIRE BANK

LOCAL MAN ARRESTED ON Hacking Charges

(Wilmington) – Donald Greene has been arrested by Interpol on computer hacking charges. Interpol claims that Greene, a computer **hacker**, planted **viruses** on unwitting **host** computers. It is thought that Greene used a method called "**piggybacking**" to gain access to computers when their owners **logged in**. This gave him access to the computer's protective firewall and allowed him to disable it.

Interpol says that Greene has done over $50 million in damage. Most of this damage was done when he used **overwriting viruses** to overwrite and destroy files on government computer systems. But Greene targeted home computers, too. For these, anti-virus software was unable to stop Greene's viruses because he included **anti-antivirus viruses** with them. These viruses essentially shut down anti-virus software. Greene is also alleged to have used **resident viruses**. These viruses **imbed** themselves in a file or disk drive. When **resident extensions** are included with them, they become a part of the computer's operating system. This makes them nearly impossible to get rid of. As with all viruses, Greene's viruses are said to have **replicated** themselves quickly to **infect** multiple computers.

Get ready!

1 Before you read the passage, talk about these questions.

1 How do computer viruses impact business?

2 What should people do when their computers are infected?

Reading

2 Read the newspaper article. Then, choose the right answers.

1 What is the article mainly about?
- **A** new types of computer virus
- **B** a man who is investigating viruses
- **C** the capture of a man who used viruses
- **D** a way to protect computers from viruses

2 Which type of virus was the most harmful?
- **A** overwriting viruses
- **B** anti-antivirus viruses
- **C** resident extensions
- **D** resident viruses

3 What can be inferred about resident viruses?
- **A** They can disable firewalls.
- **B** They aid hackers with piggybacking.
- **C** They do less damage than anti-antivirus viruses.
- **D** They do not always include resident extensions.

Vocabulary

3 Match the words (1-7) with the definitions (A-G).

1 __ piggyback
2 __ hacker
3 __ host
4 __ overwriting virus
5 __ anti-antivirus virus
6 __ resident extension
7 __ virus

- **A** a virus that attacks anti-virus software
- **B** a program file that installs itself without permission
- **C** an add-on that makes a computer virus part of the host computer's operating system
- **D** a virus that copies its code over and destroys the files of the original data
- **E** to gain unauthorized access to a computer system by using the owner's connection
- **F** a computer system that has a virus attached to it
- **G** a person who illegally exploits computer hardware and software

4 Write a word that is similar in meaning to the underlined part.

1 Email is used to <u>contaminate</u> computers with viruses.
i _ _ _ c _

2 <u>Enter identification that allows one to use a computer</u> first.
l _ _ _ _ n

3 Some viruses can <u>make copies of themselves</u>. r _ _ _ i _ _ t _

4 If the virus <u>becomes a part of something else</u>, it will be hard to delete. _ m _ _ _ s

5 <u>Viruses that become part of files but do nothing until activated</u> can go unnoticed for months.
r _ _ _ d _ _ t _ _ r _ s _ _

5 Listen and read the newspaper article again. How did Greene gain unauthorized access to other people's computers?

Listening

6 Listen to a conversation between an employee and his manager. Mark the following statements as true (T) or false (F).

1 __ The manager read the newspaper article.
2 __ Overwriting viruses destroyed the manager's files.
3 __ The manager decides to purchase new anti-virus software.

7 Listen again and complete the conversation.

Employee: Have you seen this article in today's paper? The one about the **1** _____ ?
Manager: Yes, I have. What about it?
Employee: Well, I think the IT department needs to **2** _____ _____ _____ .
Manager: Do you think our company is vulnerable to these types of attacks?
Employee: Definitely. The overwriting viruses he used could do incredible damage.
Manager: I see. How much damage are we talking about?
Employee: Greene did millions of dollars' worth of damage by **3** _____ _____ . That could happen to us.
Manager: Something like that would **4** _____ _____ _____ .
Employee: Exactly. I think these **5** _____ _____ and resident extensions are a problem, too.
Manager: Why's that?
Employee: Well, these types of viruses **6** _____ _____ in files and disk drives.

Speaking

8 With a partner, act out the roles below based on Task 7. Then, switch roles.

USE LANGUAGE SUCH AS:

Have you seen this article in today's paper?

How much damage are we talking about?

Something like that would ruin the company.

Student A: You are talking to a concerned employee. Ask Student B about:

- viruses
- potential damage

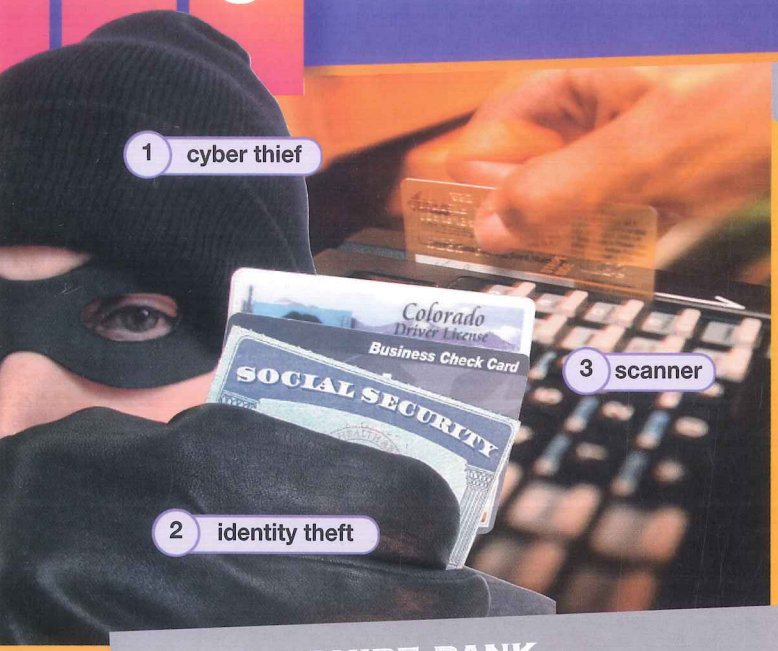
Student B: You are a concerned employee. Answer Student A's questions.

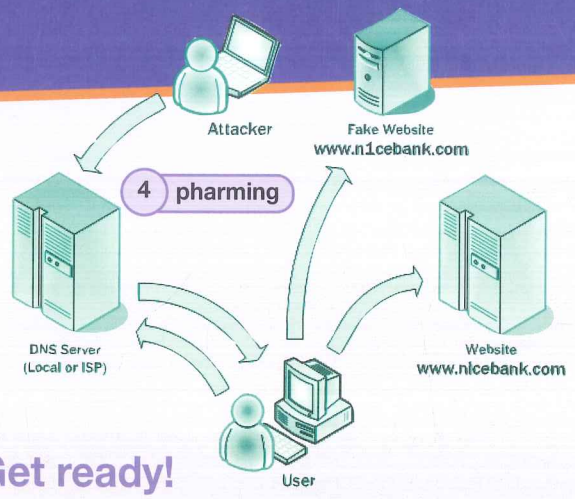
Writing

9 You are worried about computer security at work. Use the article and the conversation from Task 8 to write an email to your manager (120-150 words). Talk about:

- *Why you are concerned about computer viruses*
- *Which types of computer viruses are most dangerous*
- *How computer viruses could harm the company*

8 Identity Theft





Get ready!

1 Before you read the passage, talk about these questions.

1 How does identity theft affect businesses?
2 How can people avoid falling for phishing schemes?

Reading

2 Read the letter from a bank to its customers. Then, mark the following statements as true (T) or false (F).

1 __ Pretexting occurs on telephones.
2 __ Pharming relies on the use of card scanners.
3 __ Trojan horses are spread by pharming sites.

LINCOLNSHIRE BANK
12345 Walker Avenue Albuquerque, NM 87444
505.555.8765

Dear Valued Customer,

Recently, a series of **identity thefts** has affected our customers. Unfortunately, this led to several instances of **fraud** occurring at our bank. The best way to avoid these events is to be informed. Please take a moment to familiarize yourself with some common ways that criminals steal personal information.

Card scanning is one simple form of identity theft. This is when someone uses a card **scanner** to record the information stored on credit or debit cards. Card scanning can be used to collect passport information as well.

Email also presents opportunities for **cyber** thieves. Spam, or unsolicited emails, can contain **malware**. This malicious software includes **spyware**, **Trojan horses**, and **worms** that can infect one's computer and steal information. **Phishing** is also conducted over email. This occurs when thieves trick people into giving them information by pretending to represent a legitimate business.

Pretexting is similar to phishing but is often done over the phone. **Pharming** occurs when a hacker redirects someone to a site operated by them. The site looks legitimate and tricks people into giving away personal information.

If you believe you may be the victim of identity theft, contact us immediately. We will take steps to ensure that your assets are safe.

Vocabulary

3 Write a word that is similar in meaning to the underlined part.

1 I had a lot of <u>malicious software</u> on my old computer.
_ _ l _ _ r _

2 You have a <u>malicious virus that seems beneficial</u>.
_ _ _ _ an _ _ r _ _

3 <u>Tricking people into giving information through email</u> is a serious crime.
p _ _ _ h _ _ g

4 <u>Stealing another's personal information</u> is on the rise.
i _ _ _ _ i _ _ _ h _ f _

5 It's a <u>program that gathers personal information</u>.
_ p _ _ _ _ e

6 <u>Capturing information stored on cards</u> is a new form of stealing.
_ _ _ d _ c _ _ _ _ n _

7 <u>Using a legitimate-looking website to trick people</u> fools many people.
p _ _ _ m _ _ _

4 **Fill in the blanks with the correct words from the word bank.**

word BANK

fraud pretexting cyber worm scanner

1 The man on the phone was part of a _____ scam.
2 I got a _____ in my email and now my computer won't work.
3 The criminal used a _____ to steal the information on her card.
4 The man was arrested and went to jail for committing _____.
5 Avoid _____ crime by being safe on your computer.

5 **Listen and read the letter from a bank to its customers again. Why is Lincolnshire Bank contacting its customers?**

Listening

6 **Listen to a conversation between a customer and a bank employee. Choose the correct answers**

1 What is the customer calling about?
- **A** closing her bank account
- **B** reporting a phishing scam
- **C** flagging her account activity
- **D** changing her account information

2 What can be inferred about the woman?
- **A** She receives phishing scams often.
- **B** She has already contacted the police.
- **C** She must call the bank to get money.
- **D** She lost the money in her bank account in the scam.

7 **Listen again and complete the conversation.**

Employee: Oh! Did it ask you to give away any **1** _____ _____?
Customer: Yeah. It said that the bank **2** _____ _____ _____ my account details.
Employee: Did you email them that information?
Customer: No, I thought I should call the bank first. It seemed **3** _____.
Employee: Yes, Lincolnshire Bank would never ask for your account details via email.
Customer: That's what I thought, but the email **4** _____ _____ because it had the bank logo. It even linked to a site that looked official.
Employee: Well, some of these criminals are **5** _____ _____.

Speaking

8 **With a partner, act out the roles below based on Task 7. Then, switch roles.**

USE LANGUAGE SUCH AS:

I think I received a phishing scam.
Did you email them that information?
It even linked to a site that looked official.

Student A: You are talking to a bank employee. Ask Student B about:
- phishing scams
- what to do next
- your money's safety

Student B: You are a bank employee. Answer Student A's questions.

Writing

9 **You are a bank employee. Use the letter and the conversation from Task 8 to write about identity theft (120-150 words). Talk about:**

- *How criminals steal personal information*
- *How customers can protect their assets*

10 Anti-virus Software

FRANKLIN & JONES
security products

Web Advisor

People who want to protect their children from the dangers of the Internet use *WebAdvisor*. This software **blocks** the types of websites that you decide are unsafe. It also provides instant **web page analysis,** helping you avoid phishing and pharming scams.

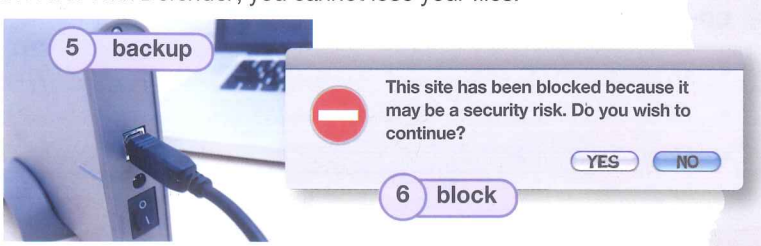
Anti-Virus Plus with RapidClean

Anti-Virus Plus is an update of our first product, *Anti-Virus*. This **anti-virus software sweeps** your computer system to **detect** viruses. All viruses are then **quarantined** so they cannot harm your computer. It's the most reliable software on the market, providing fewer **false positives** and **false negatives** than any other software.

RapidClean is a **virus removal software** that destroys the malicious software discovered by *Anti-Virus Plus*. This includes **rootkit** viruses. These viruses are designed to disguise themselves as a part of your computer's operating system. This makes them very difficult to get rid of. Most security software won't destroy rootkits but, with *RapidClean,* you can.

Defender

Defender is the ultimate **security software**. A combination of both of our software packages, *Defender* keeps your computer completely safe from **cyber criminals**. *Defender* also comes with our computer **backup** service. This means that the data on your computer's hard drive will be backed up on our company's servers. With *Defender*, you cannot lose your files.



Get ready!

1 Before you read the passage, talk about these questions.

1 What anti-virus software should a user consider buying?

2 Are there any problems with anti-virus software?

Reading

2 Read the anti-virus software advertisement. Then, choose the right answers.

1 What is the advertisement mostly about?

A damage that rootkits can cause
B kinds of web page analysis
C types of security software
D how to prevent false negatives

2 Which of the following is intended to block unsafe Internet pages?

A *Anti-Virus*
B *RapidClean*
C *WebAdvisor*
D *Anti-Virus Plus*

3 What can be inferred about false negatives and false positives?

A They aid virus detection.
B They disrupt web page analysis.
C They mark the presence of a virus.
D They occur in most anti-virus programs.

Vocabulary

3 Match the words (1-9) with the definitions (A-I).

1 __ backup
2 __ rootkit
3 __ block
4 __ sweep
5 __ quarantine
6 __ false positive
7 __ false negative
8 __ cyber criminal
9 __ web page analysis

A a person who commits illegal acts online
B to stop a file from being downloaded
C to isolate a file
D to save all information on a system
E labeling a safe file as dangerous
F labeling a dangerous file as safe
G a service that checks websites for safety
H a virus that hides itself as part of an OS
I to search a system for viruses

4 Check (✓) the sentence that uses the underlined parts correctly.

1 __ A Security software protects computers.
__ B Virus removal software infects email.

2 __ A Anti-virus software detects viruses.
__ B Phishing is malicious software.

3 __ A A rootkit notifies users of scams.
__ B Virus removal software keeps computers clean.

4 __ A It is wise to use anti-virus software.
__ B Web page analysis detects false positives.

5 Listen and read the anti-virus software advertisement again. What happens to viruses when anti-virus software sweeps the computer system?

Listening

6 Listen to a conversation between a caller and an employee. Mark the statements as true (T) or false (F).

1 __ The caller lost money in a scam.
2 __ The woman recommends software that provides web page analysis.
3 __ The Anti-Virus Plus program comes with a backup service.

7 Listen again and complete the conversation.

Caller: Well, I almost fell for a **1** _____ scam last week, so I'm worried about that.
Employee: Our WebAdvisor software can help you there. It performs **2** _____ _____ _____ or all the websites you visit.
Caller: Okay. Will it **3** _____ _____ _____ from viruses, too?
Employee: No, it won't. You'd need Anti-Virus Plus with RapidClean for that.
Caller: So I have to buy **4** _____ _____ _____?
Employee: Actually, our Defender software combines those **5** _____ _____ _____ _____.

Speaking

8 With a partner, act out the roles below based on Task 7. Then, switch roles.

USE LANGUAGE SUCH AS:

What do you think you might need?
I almost fell for a pharming scam last week.
Will it protect my computer from viruses, too?

Student A: You are talking to an anti-virus software company employee. Ask Student B about:

- software options
- software costs
- software functions

Student B: You are an anti-virus software company employee. Answer Student A's questions.

Writing

9 You are an anti-virus software company employee. Use the advertisement and the conversation from Task 8 to write out suggestions for a potential customer (120-150 words). Talk about:

- *Which types of software are available*
- *Which software you would recommend*
- *Why that particular type of software would be helpful*

Glossary

above the fold [ADJ-U4] If something on a website is **above the fold**, it is visible without scrolling down after the page has loaded.

ad product [N-COUNT-U4] An **ad product** is an advertising opportunity such as a banner that is located on a website.

ADM [N-COUNT-U12] An **ADM** (Automated Dispensing Machine) is a hospital machine that measures out specific amounts of medication into containers.

anchor [N-COUNT-U3] An **anchor** is a word, phrase, or image that includes a hyperlink and can be used for navigation.

animated GIF [N-COUNT-U4] An **animated** GIF is the combination of multiple GIF files in one advertisement to create animation.

anti-antivirus virus [N-COUNT-U7] An **anti-antivirus virus** is a computer virus that attacks and often disables anti-virus software.

anti-virus software [N-UNCOUNT-U10] **Anti-virus software** searches a computer system for computer viruses.

Apple® [N-UNCOUNT-U1] **Apple®** is the company that produces Macs® and other computing products.

applet [N-COUNT-U3] An **applet** is a small application that is embedded in a webpage.

attack [N-COUNT-U9] An **attack** is an attempt to get through computer security.

audit log [N-COUNT-U9] An **audit log** is a record of who has accessed a computer system and what actions they took.

authenticate [V-TRANS-U9] To **authenticate** something is to make sure that it is legitimate.

Automation [NOUN-UNCOUNT-U5] **Automation** is the use of programmed machines to perform tasks without human assistance.

back end [ADJ-U15] If something is **back end**, it occurs at the end of a project or job.

backdoor [N-COUNT-U9] A **backdoor** is a hidden way to get by security in computers and programs, sometimes intentionally created.

backup [V-T-U10] To **backup** is to create a duplicate copy of data that is stored on a separate hard drive or computer system to prevent the loss of the data.

bar coding [N-UNCOUNT-U12] **Bar coding** is a system in which doctors scan medicines and match them to patients.

Basecamp® [N-UNCOUNT-U6] **Basecamp®** is a program offered by the company 37 signals that organizes projects and can be accessed by multiple users simultaneously.

block [V-T-U10] To **block** something is to prevent access to it.

blog [N-U13] A **blog** is a personal website that is maintained by an individual.

bug [N-COUNT-U9] A **bug** is a flaw in a computer program.

CAD [N-UNCOUNT-U3] **CAD** (Computer-Aided Design) is the use of a computer to design things such as machines or buildings.

campaign [N-COUNT-U4] A **campaign** is an agreement between an advertising company and a client that outlines which types of advertisements are to be used, where they are to be placed, and how long they will appear.

card scanning [N-UNCOUNT-U8] **Card scanning** is the practice of capturing the personal information stored on credit cards, debit cards, or passports.

CBT [N-UNCOUNT-U3] **CBT** (Computer-Based Training) is the use of a computer to train employees.

CDN [N-COUNT-U11] A **CDN** (content delivery network) is a system of computers on a network containing copies of data that nearby users can access.

CDSS [N-UNCOUNT-U12] **CDSS** (Clinical Decision Support System) is a system that helps doctors diagnose problems. It also suggests treatment options.

centralize [V-T-U6] To **centralize** things is to bring from multiple places and make them available in one location.

click tracking [N-UNCOUNT-U4] **Click tracking** is the process of counting and keeping track of the number of clicks an advertisement receives.

click-through rate [N-COUNT-U4] The **click-through rate** is the ratio of clicked advertisements to the number of advertisements that are viewed.

cloud computing [N-UNCOUNT-U11] **Cloud computing** is use of the Internet for software and other resources.

color matching [N-UNCOUNT-U1] **Color matching** is the ability to create colors as you see them on the screen.

color-coding [N-UNCOUNT-U6] **Color-coding** is the practice of labeling things with different colors in order to tell them apart.

command line [N-COUNT-U10] A **command line** is a word or phrase that makes a computer perform an operation.

commission [N-COUNT-U4] A **commission** is the amount of money charged by a website for hosting advertisements.

component [N-COUNT-U15] A **component** is a part of something bigger.

Computer Aided Manufacturing [NOUN-UNCOUNT-U5] **Computer Aided Manufacturing** is the use of computers to control production equipment.

configure [V-T-U14] To **configure** something is to modify or arrange it so that it works in a certain way.

cover letter [N-COUNT-U14] A **cover letter** is a document that explains why a person is applying for a job and why that person believes he or she is qualified for that job.

CPC [N-UNCOUNT-U4] **CPC** (cost per click-through) is a pricing model wherein rates are charged according to how many clicks an advertisement receives.

CPM [N-UNCOUNT-U4] **CPM** (cost per 1,000 impressions) is a pricing model wherein rates are charged according to how many thousands of people view the advertisement.

CPOE [N-UNCOUNT-U12] **CPOE** (Computerized Provider Order Entry) is a system that lets a doctor fill out electronic orders.

customize [V-TRANS-U1] To **customize** something is to change it to meet your individual needs and purposes.

cyber [ADJ-U8] If something is **cyber**, it is related to computers or the Internet.

cyber criminal [N-COUNT-U10] A **cyber criminal** is a person who uses computers and the Internet to commit crimes such as identity theft and fraud.

dashboard [N-COUNT-U6] A **dashboard** is a display on a computer that shows commonly or recently used files and programs.

deployment [N-UNCOUNT-U11] **Deployment** is introducing new software into a group of hardware.

detect [V-T-U10] To **detect** something is to discover it.

duty [N-COUNT-U14] A **duty** is a task that someone is responsible for completing at a job.

DVD [N-COUNT-U3] A **DVD** (Digital Video Disc) is a storage disc that is read by a laser and can store media files such as videos and images.

editor [N-COUNT-U15] An **editor** is a program that is used to create and change files.

egress filter [N-COUNT-U13] An **egress filter** prevents harmful software from leaving a network.

EHR [N-COUNT-U12] An **EHR** (Electronic Health Record) is a computerized version of a patient's health record.

elasticity [N-COUNT-U11] **Elasticity** is a computer system's ability to gain or lose computer resources as a user sees fit.

electronic bulletin board [N-COUNT-U3] An **electronic bulletin board** is a shared file that can be used to spread information.

Embedded system [N-COUNT-U5] An **embedded system** is a combination of hardware and software designed to perform a specific task.

EMM [N-UNCOUNT-U12] **EMM** (Electronic Materials Management) is a system that helps hospitals keep track of medications and medical supplies.

Encoder [NOUN-COUNT-U5] **Encoders** are devices that convert mechanical information into electronic signals.

Ethernet network [NOUN-COUNT-U5] An **Ethernet network** is a high-speed network that links computers together.

false negative [N-COUNT-U10] A **false negative** is when anti-virus software incorrectly indicates that an infected file is clean.

false positive [N-COUNT-U10] A **false positive** is when anti-virus software incorrectly indicates that a clean file has been infected with a computer virus.

FAQ [N-COUNT-U3] A **FAQs** (Frequently Asked Questions) section is a list of common questions with their answers.

Glossary

fault tolerant [NOUN-UNCOUNT-U5] **Fault tolerant** is the ability to continue functioning even though an error has occurred.

file archiving [N-UNCOUNT-U14] **File archiving** is the process of organizing computer files in an orderly way. This lets you retrieve them easily.

firewall [N-COUNT-U9] A **firewall** is a program that restricts access on a network.

flash [N-UNCOUNT-U3] **Flash** is a media format that can be used to add videos, animations, and other media to a website.

fraud [N-UNCOUNT-U8] **Fraud** is an action that results in the loss of another person's property.

free software license [N-COUNT-U10] A **free software license** allows users to change and distribute programs.

frequency capping [N-UNCOUNT-U4] **Frequency capping** is the process of ensuring that a specific website visitor does not see the same advertisement an excessive number of times.

gallery [N-COUNT-U6] A **gallery** is a collection of icons or pictures that have been organized in a file or page.

geo targeting [N-UNCOUNT-U4] **Geo targeting** is the process of showing advertisements in geographical areas where they are relevant.

GUI [N-UNCOUNT-U15] A **GUI** (graphical user interface) is how a user interacts with a computer using images and text.

hacker [N-COUNT-U7] A **hacker** is a person who exploits computer hardware and software for their own purposes.

hang [V-T-U4] To **hang** a website is to prevent it from loading all the way.

host [N-COUNT-U7] A **host** is a computer system that has a computer virus attached to it.

HTML [N-UNCOUNT-U15] **HTML** (hypertext markup language) is a way to alter the look of text on web pages.

IaaS [N-UNCOUNT-U11] **IaaS** (infrastructure as a service) delivers such elements as servers and software over the Internet.

identity theft [N-UNCOUNT-U8] **Identity theft** is the process of stealing another person's personal information and using it for personal gain.

imbed [V-T-U7] To **imbed** is to become a part of something.

implanted microchip [N-COUNT-U12] An **implanted microchip** is a tiny device in a person's body that releases specific amounts of medicine.

infect [V-T-U7] To **infect** a computer is to contaminate it with a computer virus.

ingress filter [N-COUNT-U13] An **ingress filter** prevents harmful software from entering a network.

install [V-T-U14] To **install** something is to add a new component or program to a computer or system.

integrated circuit [N-COUNT-U5] An **integrated circuit** is a small chip that has several electronic devices attached to it.

interactive [ADJ-U15] If something is **interactive**, it will respond to a user's commands.

interface [N-COUNT-U10] An **interface** provides a way for a user to interact with a computer.

interoperability [N-UNCOUNT-U12] **Interoperability** is the ability to easily send or combine electronic health data from one system into another.

intruder [N-COUNT-U9] An **intruder** is a person or program that attempts to access a network or computer without permission.

keylogger [N-COUNT-U9] A **keylogger** is type of trojan virus that tracks what keystrokes are entered into a computer.

keyword [N-UNCOUNT-U13] A **keyword** is a term or phrase that is created to describe a webpage, so that it can be found in a search engine.

lab-on-a-chip [N-COUNT-U12] A **lab-on-a-chip** is a tiny device that holds cells and fluid that a doctor can analyze easily.

Latency [NOUN-UNCOUNT-U5] **Latency** is the time between the end of one computer communication and the start of another.

Linux® [N-UNCOUNT-U10] **Linux®** is a computer operating system.

Linux® distribution [N-COUNT-U10] A **Linux®** distribution is a collection of software for Linux®.

log in [V-I-U7] To **log in** is to provide a computer security system with the proper identification to use the computer.

Mac® [N-COUNT-U1] A **Mac®** (Macintosh) is a type of computer from Apple®.

malware [N-UNCOUNT-U8] **Malware** is any type of malicious, harmful software.

mask [V-COUNT-U13] To **mask** information is to block it from being viewed.

media player [N-COUNT-U3] A **media player** is used to play media such as videos and animations and can be embedded in a web page.

message board [N-COUNT-U6] A **message board** is an online application that displays and organizes messages from users.

microrobotic tweezers [N-COUNT-U12] **Microrobotic tweezers** are very tiny tweezers that allow a doctor to operate in extremely small or delicate areas.

Microsoft Office® [N-UNCOUNT-U10] **Microsoft Office®** is a software package that offers word processing and spreadsheet programs.

Microsoft® [N-UNCOUNT-U1] **Microsoft®** is the company that produces the Windows® operating system among other products.

MPEG [N-UNCOUNT-U3] **MPEG** is a video format that is used to compress videos into smaller files.

multimedia [N-UNCOUNT-U3] **Multimedia** is any type of media used on a webpage, including images, animations, and videos.

natural language [N-UNCOUNT-U13] **Natural language** is the use of phrases from normal speech in search engines.

open source [ADJ-U10] If software is **open source**, it can be distributed with the same basic source code.

open system [NOUN-COUNT-U5] An **open system** is a system that can use many different types of machines.

OpenOffice® [N-COUNT-U10] **OpenOffice®** is an open source software package similar to Microsoft Office®.

optic scanner [N-COUNT-U12] An **optic scanner** is a device that reads and records identification labels on medications.

OS X® [N-UNCOUNT-U1] **OS X®** is an operating system from Apple® for Mac® computers.

oscilloscope [NOUN-COUNT-U5] An **oscilloscope** is a device that allows the user to view the electrical voltage of a machine.

overwriting virus [N-COUNT-U7] An **overwriting virus** is a computer virus that copies its code over and destroys the files of the original data.

PaaS [N-UNCOUNT-U11] **PaaS** (platform as a service) is a service where users receive a computing operating system and related components over the Internet.

PACS [N-UNCOUNT-U12] **PACS** (Picture Archiving and Communications System) is a system that captures and stores medical images from different sources.

patch [N-COUNT-U9] A **patch** is a piece of code that is used to fix a flaw in a computer program.

pay as you go [N-UNCOUNT-U11] **Pay as you go** is a model for payment where customers pay according to how much they use a resource.

PC [N-COUNT-U1] A **PC** (personal computer) refers to a computer that runs a Windows® operating system.

PCI compliance [N-U13] **PCI** (payment card industry) **compliance** is upholding certain security standards for accepting credit and other payment cards.

performance tuning [N-UNCOUNT-U14] **Performance tuning** is the act of adjusting a network or system so that it works as efficiently as possible.

pharming [N-UNCOUNT-U8] **Pharming** is a process wherein thieves reroute people to a fake website that appears to be legitimate in order to trick them into giving away their personal information.

phishing [N-UNCOUNT-U8] **Phishing** is a process wherein thieves trick people into giving away their personal information through email.

PHP [N-UNCOUNT-U15] **PHP** is a scripting language that is used to create dynamic web pages.

piggyback [V-I-U7] To **piggyback** is to gain unauthorized access to a computer system by taking advantage of the owner's legitimate connection.

popup [N-COUNT-U9] A **popup** is an advertisement that suddenly appears in a new window in an Internet browser.

Glossary

prepress [N-COUNT-U1] **Prepress** is the process of preparing a document, magazine or book to be printed on a large scale.

pretexting [N-UNCOUNT-U8] **Pretexting** is a process wherein thieves trick people into giving away their personal information over the phone.

priority inversion [NOUN-COUNT-U5] A **priority inversion** is a delay that occurs in a computer system when a low-priority task is dealt with before a high-priority task.

private cloud [N-COUNT-U11] A **private cloud** is a service offered to specific users who have purchased it and is not available publicly.

processing speed [N-COUNT-U1] **Processing speed** is a measurement of how quickly a computer can perform tasks.

programmer [N-COUNT-U15] A **programmer** is someone who writes computer programs.

protocol [N-COUNT-U9] **Protocol** is a set of rules that must be followed for the sake of security.

public cloud [N-COUNT-U11] A **public cloud** is a service offered for any users on the Internet.

purge [V-T-U14] To **purge** is to delete files or data from a computer.

quarantine [V-T-U10] To **quarantine** something is to place it in complete isolation where it cannot access or infect anything else.

real time [ADJ-U6] If information is updated or shown in **real time**, it is updated or shown at the same time as new information is received.

redesign [V-COUNT-U13] To **redesign** a website is to change, add, or delete certain elements, but keep others.

redundancy [NOUN-UNCOUNT-U5] **Redundancy** is the inclusion of duplicate components in a system to maintain production when one component fails.

reliability [N-UNCOUNT-U3] **Reliability** is the probability that something will function properly and without any failures.

replicate [V-T-U7] To **replicate** is to make copies of oneself.

resident extension [N-COUNT-U7] A **resident extension** is an add-on to a computer virus that causes it to become a part of the host computer's operating system.

resident virus [N-COUNT-U7] A **resident virus** is a computer virus that becomes part of a file or disk drive but does nothing until activated by a trigger event.

résumé [N-COUNT-U14] A **résumé** is a document that shows a person's education, work history and qualifications and is used when applying for jobs.

root directory [N-COUNT-U13] The **root directory** is the main directory of a file system and is where all other branches of the system come from.

rootkit [N-COUNT-U10] A **rootkit** is a type of computer virus that is created to gain total control over a computer system by overwriting parts of its operating system.

SaaS [N-UNCOUNT-U11] **SaaS** (software as a service) is a service where users receive applications over the Internet instead of buying and installing them.

Samba® [N-UNCOUNT-U10] **Samba®** is a Linux® program that allows Linux® to act as a client on a Windows®-based network.

scalability [N-COUNT-U11] **Scalability** is a system's ability to change size as needed in a given situation.

scanner [N-COUNT-U8] A **scanner** is a device used to capture the personal information stored on credit cards, debit cards, and passports.

script [N-COUNT-U15] A **script** is a group of commands to be carried out automatically.

security [N-UNCOUNT-U9] **Security** is a the state of being safe from attack.

security software [N-UNCOUNT-U10] **Security software** is software that keeps a computer safe from threats such as computer viruses and cyber crimes.

Skype™ [N-UNCOUNT-U6] **Skype™** is software than enables people to call one another using computers rather than telephones.

spyware [N-UNCOUNT-U8] **Spyware** is a type of computer program that gathers someone's personal information without their knowledge.

sweep [V-T-U10] To **sweep** a computer system is to search it for computer viruses.

system administrator [N-COUNT-U14] A **system administrator** is a person whose job is to manage and maintain a computer system.

TCP/IP protocol [N-COUNT-U10] A **TCP/IP protocol** is a set of rules that determine how a computer connects to a network or the Internet.

technical [ADJ-U14] Something that is **technical** involves information from the areas of science and industry.

telecommute [V-I-U6] To **telecommute** is to work at a location away from a main office by using technology.

terminal [N-COUNT-U3] A **terminal** is a computer that allows users to input command lines.

toggle [V-I-U6] To **toggle** is to move from one file or setting to another.

tour [N-COUNT-U6] A **tour** is a series of web pages that offer information about a product or site.

Trojan horse [N-COUNT-U8] A **Trojan horse** is a computer program that appears to be beneficial but is actually malicious.

troubleshooting [N-UNCOUNT-U14] **Troubleshooting** refers to the process of identifying and fixing technical problems.

trust logo [N-COUNT-U13] A **trust logo** is a seal that appears on a web page and means that the security of information on that page is assured by an accredited company.

typography [N-COUNT-U1] **Typography** is a computer's use of typefaces.

Ubuntu® [N-UNCOUNT-U10] **Ubuntu®** is an operating system based on Linux.

Unix® [N-UNCOUNT-U14] **Unix®** is a specific type of computer operating system. Servers often use this operating system.

user-friendly [ADJ-U15] If something is **user-friendly**, it is easy to access and use.

utility computing [N-UNCOUNT-U11] **Utility computing** is a way of selling computer use as a metered service like water and electricity.

vendor [N-COUNT-U10] A **vendor** is someone who distributes programs and utilities.

video subsystem [N-COUNT-U1] The **video subsystem** is the part of the computer that presents the visuals on the screen.

virus [N-COUNT-U7] A **virus** is a harmful computer program file that takes up residence in a computer without the owner's knowledge or permission.

virus removal software [N-UNCOUNT-U10] **Virus removal software** destroys computer viruses, removing them from the files they have infected.

visual basic [N-UNCOUNT-U15] **Visual basic** is a simple programming language with a graphical component.

web developer [N-COUNT-U15] A **web developer** is someone who builds a website.

web page analysis [N-UNCOUNT-U10] **Web page analysis** is a process wherein the security of a website is determined in order to help computer users know whether or not the website is safe.

web portal [N-COUNT-U13] A **web portal** is a webpage that displays information gathered from several different places.

web-based [AD-U6] If something is **web-based**, it is available on the Internet instead of on a disk.

Windows® [N-UNCOUNT-U1] **Windows®** is the operating system created by Microsoft® that is used on many PCs.

worm [N-COUNT-U8] A **worm** is a type of malicious software that replicates itself through emails.

XML [N-COUNT-U15] **XML** (extensive markup language) is a programming language that is used to encode web pages.

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